

Calcium transfer from the ER to other organelles for optimal signaling in *Toxoplasma gondii*

Zhu-Hong Li, Beejan Asady, Le Chang, Miryam Andrea Hortua Triana, Catherine Li, Isabelle Coppens, Silvia NJ Moreno 

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States • Department of Molecular Microbiology and Immunology, Johns Hopkins Bloomberg School of Public Health, Baltimore, United States • Department of Cellular Biology, University of Georgia, Athens, United States

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This **important** study shows that calcium stores in the endoplasmic reticulum of the parasitic protozoan, *Toxoplasma gondii* play a major role in buffering calcium levels in the cytosol as well as other organelles such as the mitochondrion. Advanced imaging techniques, including use of genetically encoded calcium indicators provide **compelling** evidence for the role of the SERCA-Ca²⁺ ATPase pump in regulating organellar calcium levels. However, it remains unclear whether intra-organellar calcium transport occurs via ER-mitochondria membrane contact sites or other mechanisms. This work will be of interest to cell and molecular biologists interested in calcium signalling in divergent eukaryotes.

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Abstract

Ca²⁺ signaling in cells begins with the opening of Ca²⁺ channels in either the plasma membrane (PM) or endoplasmic reticulum (ER), leading to a sharp increase of the physiologically low (<100 nM) cytosolic Ca²⁺ level. The temporal and spatial regulation of Ca²⁺ is crucial for the precise activation of key biological processes. In the apicomplexan parasite *Toxoplasma gondii*, which infects approximately one-third of the global population, Ca²⁺ signaling governs essential aspects of the parasite's infection cycle. *T. gondii* relies on Ca²⁺ signals to regulate pathogenic traits, with several Ca²⁺-signaling components playing critical roles. Ca²⁺ entry from the extracellular environment has been demonstrated in *T. gondii* for both, extracellular parasites, exposed to high Ca²⁺, and intracellular parasites, which acquire Ca²⁺ from host cells during host Ca²⁺ signaling events. Active egress, an essential step of the parasite's infection cycle, is preceded by a large increase in cytosolic Ca²⁺, most likely initiated by release from intracellular stores. However, extracellular Ca²⁺ is also necessary to reach a cytosolic Ca²⁺ threshold required for timely egress. In this study, we investigated the mechanism of Ca²⁺ intracellular store replenishment and identified a central role for the SERCA-Ca²⁺-ATPase in maintaining Ca²⁺ homeostasis not only within the ER but also in other

organelles. We demonstrate mitochondrial Ca^{2+} uptake, which occurs by transfer of Ca^{2+} from the ER, likely through membrane contact sites. Our findings suggest that the *T. gondii* ER plays a key role in sequestering and redistributing Ca^{2+} to intracellular organelles following Ca^{2+} influx at the PM.

Highlights

The *T. gondii* ER takes up Ca^{2+} that enters the cytosol from the extracellular milieu.

Filling of acidic stores in *T. gondii* depends on ER Ca^{2+} content.

The mitochondrion of *T. gondii* has no direct access to extracellular Ca^{2+} but can take it up via transfer from the ER and/or acidic stores.

The absence of SERCA activity results in reduced Ca^{2+} levels in the ER as well as in other organelles

Introduction

Toxoplasma gondii is an intracellular parasite from the Apicomplexan Phylum that infects approximately one third of the world population (Weiss and Dubey, 2009 [↗](#)). During the initial infection, *T. gondii* undergoes multiple rounds of a lytic cycle, which consists of host cell invasion, replication within a parasitophorous vacuole (PV), exit from the host cell causing its lysis followed by reinvasion of new host cells (Black and Boothroyd, 2000 [↗](#); Blader et al., 2015 [↗](#)). Cytosolic Ca^{2+} ($[\text{Ca}^{2+}]_c$) fluctuations precede the activation of several key steps of the *T. gondii* lytic cycle like motility, attachment, invasion, and egress (Lourido and Moreno, 2015 [↗](#); Hortua Triana et al., 2018 [↗](#)). Egress from the host cell is an essential step for the infection cycle of *T. gondii* (Bisio and Soldati-Favre, 2019 [↗](#)) and it was shown that it is preceded by a cytosolic Ca^{2+} increase (Endo et al., 1982 [↗](#); Borges-Pereira et al., 2015 [↗](#)). Extracellular Ca^{2+} entry was demonstrated in extracellular (Pace et al., 2014 [↗](#); Hortua Triana et al., 2024 [↗](#)) and intracellular replicating tachyzoites (Vella et al., 2021 [↗](#)). This activity was highly regulated and work from our lab revealed that a TRP-like activity was involved (Marquez-Nogueras et al., 2021 [↗](#)).

Ca^{2+} signaling is part of the signaling pathways that regulate a large number of cellular functions (Clapham, 2007 [↗](#)). All cells express a variety of channels, transporters, and Ca^{2+} pumps, located at the plasma membrane (PM) and/or intracellular organelles (endoplasmic reticulum (ER), acidic stores, and mitochondria) that regulate/control the concentration of cytosolic Ca^{2+} . However, an elevated cytosolic Ca^{2+} concentration sustained for prolonged periods is toxic to cells and may result in their death (Bootman and Bultynck, 2020 [↗](#)).

In *T. gondii*, both Ca^{2+} entry through the plasma membrane and release from intracellular stores like the ER may initiate a cascade of signaling events important for the stimulation of the biological steps of the parasite lytic cycle (Lourido and Moreno, 2015 [↗](#); Hortua Triana et al., 2018 [↗](#)). Ca^{2+} oscillations were observed in motile parasites loaded with fluorescent Ca^{2+} indicators, (Lovett and Sibley, 2003 [↗](#)) as well as expressing Genetically Encoded Calcium Indicators (GECIs) (Borges-Pereira et al., 2015 [↗](#)). The significance of Ca^{2+} signals during all stages of the lytic cycle has been demonstrated, but little is known about the mechanism by which intracellular stores contribute to cytosolic Ca^{2+} signals and downstream regulation.

The ER, an exclusive organellar feature of eukaryotic cells, is the main store for Ca^{2+} in most eukaryotic cells. It has been proposed that the ER is functionally heterogeneous with Ca^{2+} binding proteins and Ca^{2+} pumps and channels having a nonuniform distribution resulting in the presence

of distinct subdomains within the organelle (Papp et al., 2003 [↗](#)). The ER in mammalian cells facilitates Ca^{2+} tunneling through its lumen as a mechanism of delivering Ca^{2+} to targeted sites without activating inappropriate processes in the cell cytosol (Petersen et al., 2017 [↗](#)). In addition, the ER is ubiquitously distributed and is in close contact with all cellular organelles and the plasma membrane (PM) (Spang, 2018 [↗](#)). Over the past decade a new paradigm has emerged that seeks to decipher how subcellular organelles communicate with each other in order to coordinate activities and efficiently distribute ions and lipids within the cell. Numerous observations have highlighted the presence of tight, stable and yet non-fusogenic associations between organellar membranes which have since become known as membrane contact sites (MCSs) (Phillips and Voeltz, 2016 [↗](#)).

The secretory pathway of *T. gondii* is organized in a highly polarized manner with the ER being an extension of the nuclear envelope (Hager et al., 1999 [↗](#); Tomavo et al., 2013 [↗](#)). The ER at the apical surface of the nuclear envelope continues with the Golgi stacks and then the secretory organelles, micronemes and rhoptries, which are unique to the apicomplexan phylum (Hager et al., 1999 [↗](#)). These organelles perform important functions relevant to a successful lytic cycle like host cell attachment, invasion, and establishment of the parasitophorous vacuole (PV). Cytoplasmic Ca^{2+} increases, due to release from the ER, have been reported to initiate responses like microneme secretion (Carruthers and Sibley, 1999 [↗](#); Nagamune et al., 2007b [↗](#)), conoid extrusion (Del Carmen et al., 2009 [↗](#)), invasion (Vieira and Moreno, 2000 [↗](#); Lovett and Sibley, 2003 [↗](#)) and egress (Arrizabalaga and Boothroyd, 2004 [↗](#)). These responses require precise spatiotemporal control of diverse targets and suggests the presence of distinct systems to deliver Ca^{2+} to specific locations rather than allowing global increases, which would activate unnecessary and potentially detrimental signaling events (Huet and Moreno, 2023 [↗](#)).

In order to concentrate Ca^{2+} ions, the ER utilizes a SERCA- Ca^{2+} -ATPase, a transmembrane P-type ATPase, that couples ATP hydrolysis to the transport of ions across biological membranes and against a concentration gradient. SERCA pumps can be inhibited by various inhibitors including the very potent and highly specific thapsigargin (TG) (Thastrup et al., 1990 [↗](#); Sagara and Inesi, 1991 [↗](#)).

T. gondii expresses a SERCA Ca^{2+} -ATPase (TgSERCA), which possesses conserved SERCA domains, Ca^{2+} -binding sites, and residues required for ATP hydrolysis (Nagamune et al., 2007a [↗](#)). The function of TgSERCA was determined by rescue experiments of yeast cells defective in Ca^{2+} -ATPases and by its specific inhibition by TG (Nagamune et al., 2007a [↗](#)). TgSERCA was mainly localized to the ER of *T. gondii* but also showed a distinct distribution in extracellular parasites, where the protein was partially found in ER vesicles in the apical region near micronemes (Nagamune et al., 2007a [↗](#)). This distribution pattern was different to the one obtained with the transient transfection of GFP-HDEL (an ER marker), which was retained near the nuclear envelope, suggesting an uneven distribution of ER markers in extracellular parasites. The authors suggested that this distribution to the apical end may be important for rapid release and effective recovery of cytosolic Ca^{2+} , events that likely govern both motility and microneme secretion (Nagamune et al., 2007a [↗](#); Nagamune et al., 2007b [↗](#)).

In this work, we investigate the role of the ER in intracellular calcium handling in *Toxoplasma gondii*. Specifically, we explore how the ER contributes to calcium uptake following extracellular influx and how it facilitates calcium redistribution to other organelles. Using genetic and pharmacological tools, we examine the activity of the SERCA Ca^{2+} -ATPase and its role in coordinating calcium dynamics. Our findings support a model in which the ER acts as a central hub for calcium buffering and trafficking, driven by the high calcium affinity of TgSERCA.

Results

The ER sequesters and redistributes Ca^{2+} to other organelles following PM influx

We initially designed experiments to track the destination of Ca^{2+} taken up by extracellular tachyzoites from the extracellular milieu into their cytosol. Tachyzoites were loaded with the ratiometric Ca^{2+} indicator Fura-2 (**Fig. 1A**) and incubated in Ringer buffer containing 100 μM EGTA to chelate extracellular Ca^{2+} and prevent further uptake. Under these conditions, we investigated the release of intracellular Ca^{2+} pools by various pharmacological agents.

When thapsigargin (TG), an inhibitor of SERCA, was added, it blocked the re-uptake of Ca^{2+} into the ER and unmasked the passive leak of Ca^{2+} from the ER into the cytosol, resulting in a cytosolic Ca^{2+} increase of approximately 160 nM (**Fig. 1B-C**). We compared this response to the effect of ionomycin (IO), a $\text{Ca}^{2+}/\text{H}^{+}$ ionophore which acts on neutral intracellular Ca^{2+} stores, inducing depletion of Ca^{2+} , mainly from the ER (Smith et al., 1989). Exposure of *T. gondii* to 1 μM IO caused a cytosolic increase of approximately 700-1100 nM Ca^{2+} (**Fig. 1C**). We next tested another SERCA inhibitor, cyclopiazonic acid (CPA), which is structurally unrelated to TG and with a different mode of action (Inesi and Sagara, 1994). CPA induced a smaller increase in cytosolic Ca^{2+} compared to TG, possibly indicating less efficacy toward TgSERCA (**Fig. S1A-B**). Exposure to TG before CPA abolished the effect of CPA, whereas exposure to CPA did not prevent the effect of TG. (**Fig. S1B-C**). Although TG does a better job at depleting the ER of Ca^{2+} in intact parasites, the resulting increase of cytosolic Ca^{2+} after adding TG is modest compared to the response of IO (**Fig. 1C**). This result could reflect the slow kinetics of Ca^{2+} leak from the ER, allowing other buffering and transport mechanisms to mitigate the phenomenon. Alternatively, it may indicate the duration after TG treatment allowing time to complete store depletion. As shown in **Fig. 1B-C**, residual Ca^{2+} remains in the stores after TG treatment, and the TG-induced phenomenon does not return to baseline, suggesting that the leak remains active.

Next, we tested the effect of Ca^{2+} entry in the filling of intracellular stores by measuring the cytosolic Ca^{2+} increases in response to inhibitors after pre-exposing the parasite suspension to extracellular Ca^{2+} (**Fig. 1D-G**, darker traces). Addition of 1.8 mM Ca^{2+} caused a cytosolic increase due to influx through the plasma membrane (**Fig. 1D**, dark blue trace) (Pace et al., 2014). A substantial portion of the entering Ca^{2+} appeared to be rapidly sequestered by the ER, as evidenced by the significantly greater TG-induced response in parasites previously exposed to extracellular Ca^{2+} (compare light and dark blue traces) (**Fig. 1D**). We next tested the lysosomotropic agent glycyl-L-phenylalanine-naphthylamide (GPN), which primarily mobilizes Ca^{2+} from acidic organelles (Haller et al., 1996; Lloyd-Evans et al., 2008; Miranda et al., 2010; Yuan et al., 2021) and observed a similar pattern (**Fig. 1E**). The increase in cytosolic Ca^{2+} following GPN addition was markedly greater in parasites previously exposed to extracellular Ca^{2+} than in those that had not been exposed (**Fig. 1E**, compare light and dark blue traces). We previously proposed that GPN may act on the lysosome-like Plant-Like Vacuolar Compartment (PLVAC), a dynamic acidic organelle involved in calcium storage, and the processing of secretory proteins (Miranda et al., 2010; Stasic et al., 2022). The increase in cytosolic Ca^{2+} in response to the addition of nigericin (which acts on acidic stores) or CCCP (which likely targets mitochondria) was also greater in cells previously loaded with extracellular Ca^{2+} (**Fig. 1F-G**, dark purple and orange traces, respectively). These data indicate that the ER, mitochondrion, PLVAC, and other acidic stores release more calcium into the cytosol of *T. gondii* tachyzoites following exposure to extracellular calcium, which stimulates its influx through the plasma

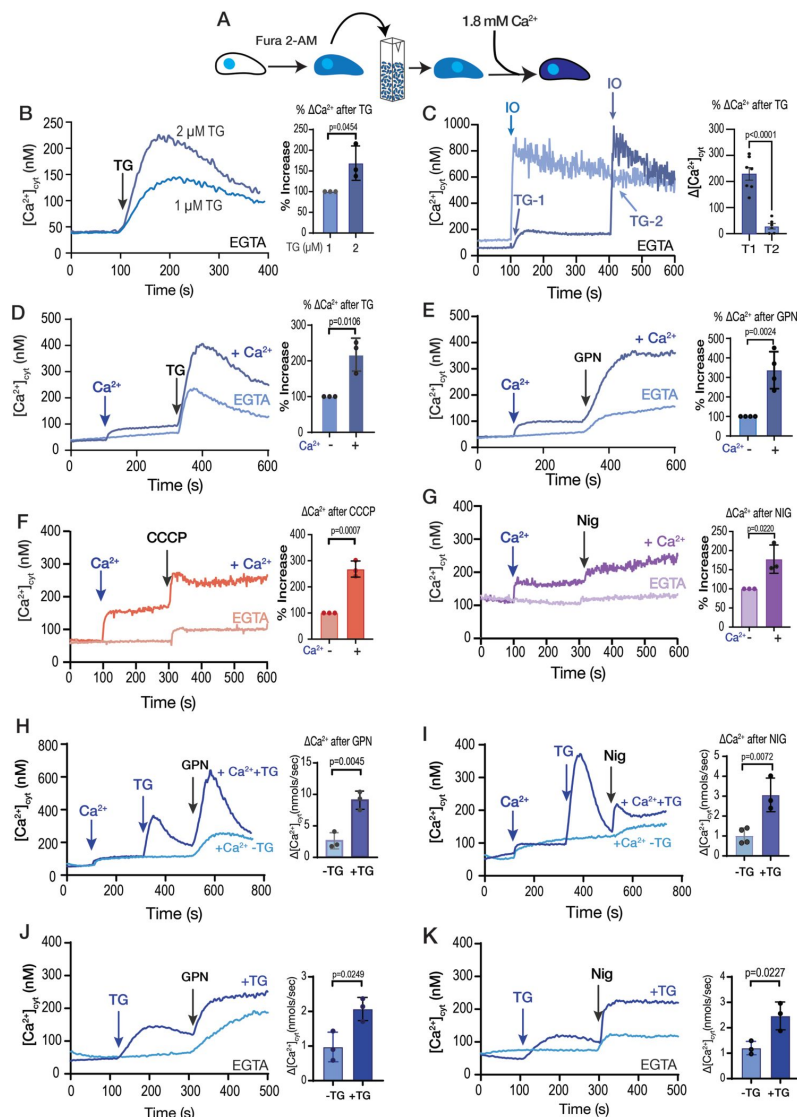


Figure 1

The role of extracellular calcium in the filling of intracellular stores.

A, scheme depicting the Fura-2-AM loading and the experimental set-up. **B**, *T. gondii* tachyzoites loaded with Fura-2 were in suspension in Ringer buffer with 100 μ M EGTA. Thapsigargin (TG) was added at 100 seconds, at two different concentrations (1 and 2 μ M). **C**, Same conditions as in **B**. 1 μ M Thapsigargin (TG) was added at 100 sec for T1 and 400 sec for T2. 1 μ M ionomycin (IO) was added at 400 sec for trace 1 (T1) and at 100 sec for trace 2 (T2). Bar graph shows Ca^{2+} increase after adding TG before (T1) and after IO (T2). **D**, Same conditions as in **B**. 1.8 mM CaCl_2 was added at 100 sec, followed by 2 μ M TG at 300 sec (dark blue trace). The light blue trace shows the same experiment without the addition of CaCl_2 . **E**, similar to **D** but using 40 μ M GPN instead of TG. **F**, same experimental set-up to the one shown in **D** but using the mitochondrial uncoupler CCCP. **G**, same experimental set-up to the one shown in **D** but adding the potassium ionophore nigericin (Nig), 10 μ M. The quantification for **D**, **E**, **F**, and **G** shows the % increase of cytoplasmic calcium compared with the same condition without calcium. **H**, 1.8 mM CaCl_2 was added at 100 sec, 1 μ M TG was added at 300 sec followed by 40 μ M GPN at 500 sec. The light blue trace shows the same experiment without the addition of TG. The quantification shows the Ca^{2+} increase after adding GPN $\pm$ previous addition of TG. **I**, Identical experiment to **H** but instead using 10 μ M Nig at 500 sec. The quantification shows the Ca^{2+} increase after adding Nig $\pm$ previous addition of TG. **J**, 1 μ M TG was added at 100 sec followed by 40 μ M GPN at 300 sec. The light blue trace shows the same experiment without the addition of TG. The buffer contains 100 μ M EGTA. Quantification shows Ca^{2+} increase after adding GPN $\pm$ previous addition of TG. **K**, Similar conditions to **J** but with 10 μ M Nig at 300 sec instead. Data are presented as mean $\pm$ SD for all comparisons. *p* value: unpaired two tailed t test performed in all comparisons.

membrane. The ER displayed high capacity to access a large portion of extracellular Ca^{2+} , with TG producing close to $\sim 300\text{--}400$ nM of Ca^{2+} increase after pre-exposure to Ca^{2+} and only $\sim 150\text{--}200$ nM Ca^{2+} without Ca^{2+} pre-exposure (**Fig. 1D** [↗](#)).

We next aimed to understand how other compartments are replenished with Ca^{2+} , given that the ER appears to be particularly effective at taking up Ca^{2+} from the cytosol. We designed an experiment where parasites were first loaded with Ca^{2+} , followed by inhibition of SERCA using TG. This inhibition prevents ER Ca^{2+} uptake, allowing Ca^{2+} to accumulate on the cytosolic side of the ER membrane. Under these conditions, we added agonists such as GPN or nigericin following the addition of TG. As shown in **Fig. 1H** [↗](#), Ca^{2+} was first added to load intracellular stores, followed by TG to induce ER Ca^{2+} leakage, and then GPN to trigger Ca^{2+} release from acidic stores. Comparison of GPN-induced cytosolic Ca^{2+} signals with and without TG pre-treatment revealed a significantly greater response in the TG condition. A similar enhancement was observed for the nigericin-induced response following TG treatment (**Fig. 1I** [↗](#)).

These results support our hypothesis that extracellular Ca^{2+} is primarily taken up by the ER and subsequently redistributed to other organelles. Importantly, Ca^{2+} was added prior to TG to allow store loading, and TG treatment then permitted ER Ca^{2+} leakage, facilitating Ca^{2+} transfer to other compartments.

However, we considered the possibility that the enhanced responses to GPN or nigericin could be due to increased PM Ca^{2+} influx triggered by elevated cytosolic Ca^{2+} . To test this, we repeated the experiments in the absence of extracellular Ca^{2+} (**Fig. 1J–K** [↗](#)). Notably, prior addition of TG resulted in an enhanced cytosolic Ca^{2+} response to both GPN and nigericin. These results further support the notion that Ca^{2+} can be transferred from the ER to other intracellular stores independently of extracellular Ca^{2+} influx.

We also performed an additional experiment in which SERCA was inhibited with TG prior to Ca^{2+} addition. We then quantified the subsequent GPN response in conditions with and without TG preincubation and observed a significant increase in the TG-treated group (**Fig. S2A** [↗](#)). This result suggests that, under non-physiological conditions where SERCA is blocked, the PLVAC may take up Ca^{2+} directly from the cytosol. However, this is unlikely to occur under normal conditions, as functional SERCA likely has a higher affinity for Ca^{2+} and would sequester it limiting its availability to other compartments.

In summary, pre-exposure of *T. gondii* to physiological levels of extracellular Ca^{2+} markedly enhanced the capacity of the ER, mitochondria, and acidic stores to release calcium, with the ER and GPN-sensitive stores exhibiting the most pronounced responses.

Ca^{2+} uptake by the SERCA- Ca^{2+} ATPase in permeabilized tachyzoites

The previous results highlight the central role of the ER in taking up Ca^{2+} from the cytoplasm following an influx from the extracellular milieu. We propose that this ER uptake, essential for maintaining Ca^{2+} store levels, is driven by the high Ca^{2+} affinity of TgSERCA. As a key mechanism, SERCA enables the ER to sustain its Ca^{2+} concentration despite the constitutive and passive leakage of Ca^{2+} from the ER into the cytosol ([Camello et al., 2002](#) [↗](#)).

To characterize the activity of TgSERCA *in situ*, we adapted a protocol to directly measure Ca^{2+} uptake by the stores in which TgSERCA localizes (ER and Golgi apparatus) ([Calixto et al., 2025](#) [↗](#)). This approach, which has been widely used in mammalian cells to assess Ca^{2+} release from the ER, employs the low-affinity Ca^{2+} indicator Mag-Fluo-4 ($K_d \sim 22$ μM) ([Rossi and Taylor, 2020](#) [↗](#)). The cytosolic concentration of Ca^{2+} in *T. gondii* is approximately 70 nM ([Moreno and Zhong, 1996](#) [↗](#)), which is well below the detection threshold of Mag-Fluo-4. To facilitate loading into organelles, we

incubated parasites for an extended period with higher concentrations of Mag-Fluo-4-AM, promoting its compartmentalization into intracellular stores. Following incubation, parasites were washed and treated with a low concentration of digitonin, which selectively permeabilizes the plasma membrane while preserving the integrity of organellar membranes. Under these conditions, the parasites retained the Ca^{2+} indicator within their organelles (**Fig. 2A**). We next assessed the capacity of these permeabilized parasites to take up Ca^{2+} . Since the activity of SERCA depends on MgATP (**Fig. 2B**), we added this substrate in the presence of defined Ca^{2+} concentrations calculated using the MaxChelator program (Bers et al., 1994). Under these conditions (free calcium ranging from 55–880 nM and MgATP at concentrations of 25–500 μM), we observed consistent and reproducible Ca^{2+} uptake, as shown in **Fig. 2B–C**. We selected 220 nM Ca^{2+} for our study because this concentration approximates physiological cytosolic fluctuations and supports detectable Ca^{2+} uptake. Additionally, this concentration of Ca^{2+} has been used in previous studies of mammalian SERCA (Rossi and Taylor, 2020). Validation that this activity is mediated by TgSERCA is demonstrated by the addition of TG, which inhibits SERCA allowing Ca^{2+} leakage from the organelle (**Fig. 2D**). Although the Ca^{2+} released after adding TG appears modest, consistent with the slow leak characteristics of ER calcium, the high K_d (22 μM) of the indicator implies that even small decreases in fluorescence signal, represent significant Ca^{2+} efflux. In contrast, IO at 1 μM caused a more pronounced Ca^{2+} release, lowering the Ca^{2+} concentration below the baseline level. This effect is likely due to IO targeting multiple intracellular compartments in addition to the ER, as well as the fundamental difference in mechanisms: IO acts as an ionophore, directly facilitating Ca^{2+} efflux across membranes, whereas TG inhibits SERCA, resulting in Ca^{2+} release through the ER's natural leak pathway (**Fig. 2E**).

We used the Mag-Fluo-4 assay to directly compare the inhibitory effects of CPA and TG (**Fig. S3**). Under the conditions of the Mag-Fluo-4 assay, using digitonin-permeabilized parasites, both inhibitors produced comparable levels of Ca^{2+} efflux suggesting that at the concentrations used both inhibited SERCA and the efflux rate corresponds to the intrinsic ER leak mechanism (**Fig. S3A–C**). This finding suggests that CPA may be less effective at inhibiting SERCA in intact parasites, possibly due to its reversibility and partial dissociation over time, allowing residual Ca^{2+} reuptake into the ER and resulting in a smaller cytosolic Ca^{2+} increase compared to TG.

In summary, these results demonstrate that the activity of TgSERCA in *T. gondii* tachyzoites can be measured *in situ* using permeabilized parasites loaded with the low-affinity Ca^{2+} indicator Mag-Fluo-4. This activity is MgATP-dependent and both TG and CPA can inhibit TgSERCA activity, leading to leakage of the accumulated Ca^{2+} . The larger effect of IO compared to TG is likely due to differences in their mechanisms of action.

TgSERCA and the *T. gondii* lytic cycle

To investigate the role of TgSERCA (TGGT1_230420) in the biology of *T. gondii* we generated conditional knockout parasites (*ΔTgSERCA*), based on the gene's predicted essentiality (fitness score -5.44) (Sidik et al., 2016). A tetracycline regulatable element was inserted at the 5' end of the *TgSERCA* gene to control its expression with anhydrotetracycline (ATc) (Sheiner et al., 2011). In addition, we endogenously tagged TgSERCA with a C-terminal 3xHA epitope and generated clonal lines of both *ΔTgSERCA* and *ΔTgSERCA-3HA* (**Fig. 3A**).

With the aim to detect the protein, we generated a guinea pig polyclonal antibody against the phosphorylation (P) and nucleotide-binding (N) domains of TgSERCA, which was affinity-purified and validated by Western blotting (**Fig. 3B**) and IFAs (**Fig. S4**). Colocalization of TgSERCA with HA epitope was confirmed by IFA. Although the signals from the anti-HA and anti-TgSERCA did not completely overlap, both were lost in the *ΔTgSERCA* mutant grown in the presence of ATc (**Fig. S4**, +ATc). The partial colocalization may reflect differences in antibody accessibility or that the two antibodies recognize distinct regions of the protein. Both Western blots and IFAs confirmed that TgSERCA expression is tightly regulated by ATc and becomes undetectable after 2.5 days in culture (**Fig. 3B** and **S4**). Growth of the *ΔTgSERCA* mutant was severely impaired in the

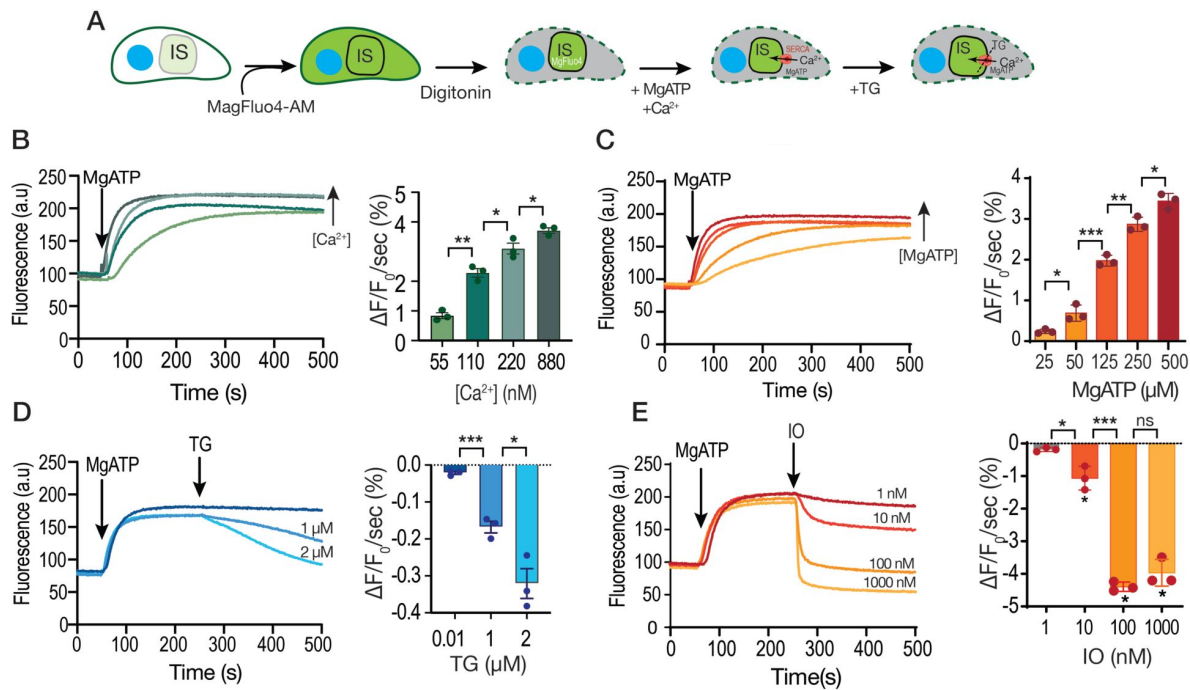


Figure 2

Ca²⁺ uptake by intracellular stores.

A, Scheme showing the loading with Mag-Fluo-4 AM followed by permeabilization with digitonin of a *T. gondii* tachyzoite (RH parental strain) suspension (IS, intracellular store). **B**, Fluorescence measurements (see Materials and Methods for specifics) of the suspension of parasites loaded with Mag-Fluo-4. MgATP (500 μ M), the SERCA substrate was added at 50 seconds. The bar graph shows the quantification of the slope of the increase in fluorescence after adding MgATP. The concentration of free calcium was varied, and it is indicated. The calculation of free calcium was done using MaxChelator. **C**, similar experimental set-up to the one shown in B with 220 nM free Ca²⁺, with varied concentrations of MgATP as indicated in the bar graph, which shows the quantification of the slope of fluorescence increase after adding MgATP. **D**, experiment was done with 500 μ M MgATP and 220 nM free Ca²⁺. Thapsigargin (TG) was added to inhibit SERCA causing calcium to be released from the store. The concentrations used are indicated. The bar graph shows the negative slope after the addition of TG. **E**, similar to D, but adding various concentrations of ionomycin (IO). The concentrations used are indicated and the slopes were measured after the addition of IO. Data are presented as mean $\pm$ SD for B-D. *p* value: unpaired two tailed t test performed in all comparisons. ns, not significant, *p* > 0.05. *, *p* $\leq$ 0.05. **, *p* $\leq$ 0.01. ***, *p* $\leq$ 0.001. ****, *p* $\leq$ 0.0001

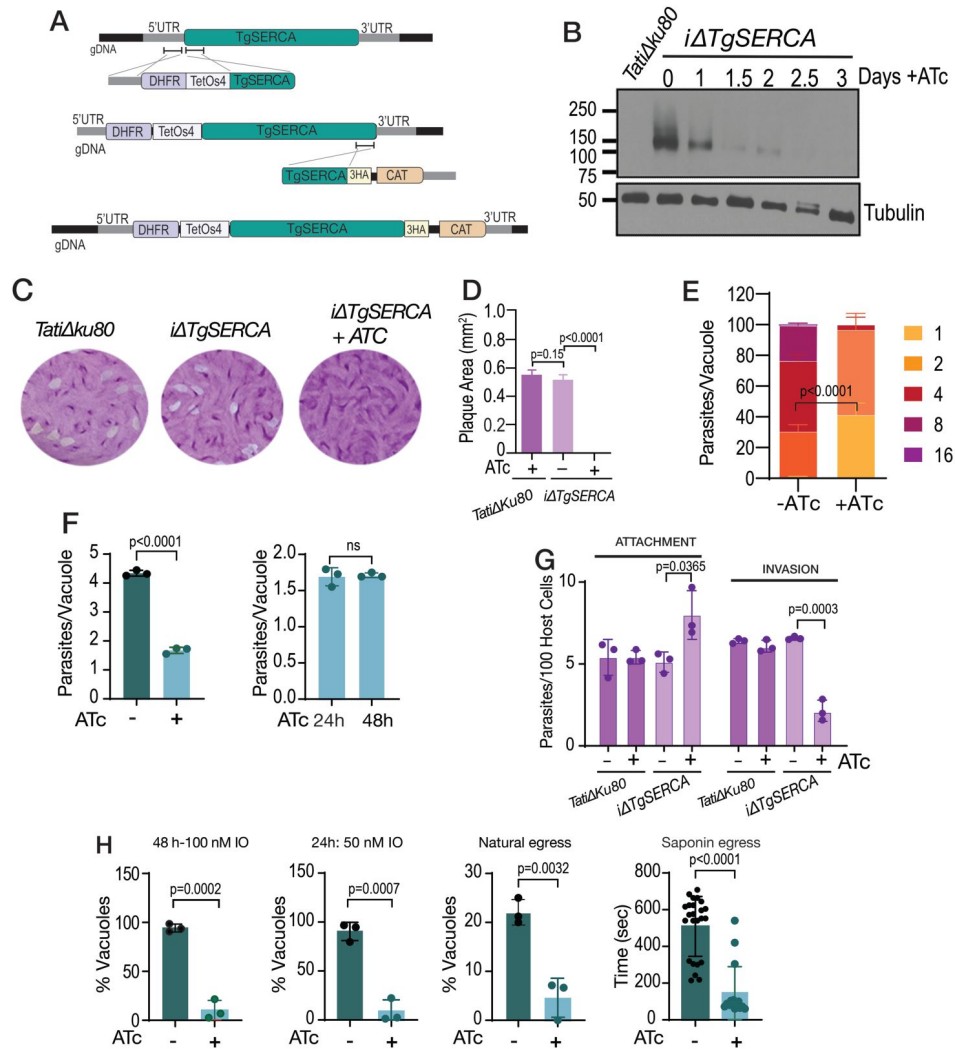


Figure 3

The sarcoplasmic-endoplasmic reticulum calcium ATPase (SERCA) is essential for the *T. gondii* lytic cycle.

A, Scheme showing the strategy used for generating conditional knock outs of TgSERCA by promoter insertion and regulation by 0.5 μg/ml Anhydrotetracyclin (ATc). The resulting mutants were named *iΔTgSERCA* or *iΔTgSERCA-3HA* (C-terminally HA-tagged). DHFR, dihydrofolate reductase gene (pyrimethamine selection); CAT, chloramphenicol acetyltransferase gene (chloramphenicol selection). **B**, Western blots of *iΔTgSERCA-3HA* parasites grown ± ATc. TgSERCA expression was detected using an anti-HA antibody, showing reduced levels with ATc treatment. **C**, plaque assays comparing the growth of *iΔTgSERCA* tachyzoites (150 parasites/well) cultured ± 0.5 μg/ml ATc for 8 days. Plaques formed by the parental *TatiΔku80* strain are shown for comparison. **D**, quantification of the size of the plaques presented in C. **E**, Replication assay using the *iΔTgSERCA-RFP* mutant. The number of parasites per parasitophorous vacuole (PV) was quantified 24 h post-infection of fibroblast cells and compared between parasites grown ± 0.5 μg/ml ATc. **F**, Average number of parasites per PV counted at 24 h after the initial infection. The graph to the right shows the number of parasites per PV of the *iΔTgSERCA* (+ATc) for 24 or 48 hours after the initial infection. **G**, Invasion assay of the *iΔTgSERCA* mutant following 24 h of ATc treatment, performed using the red-green assay described in Materials and Methods. **H**, Egress assays with fibroblast monolayers infected with *iΔTgSERCA-RFP* parasites for 24 or 48 h. Egress was triggered with ionomycin (IO; 100 nM or 50 nM) or saponin (0.01%). Natural egress was monitored following treatment with 1 μM compound 1 as described in the Methods section. % Vacuoles: 100 X Vacuoles egressed/total vacuoles. Data (D, E, F, G, H) are presented as mean from at least three biological replicates ± SD. Statistical significance was assessed using an unpaired two-tailed t-test.

presence of ATc, as assessed by plaque assays (**Fig. 3C–D**). In this assay, parasites undergo successive rounds of invasion, replication, and egress, leading to host cell lysis and the formation of plaques on confluent monolayers. Downregulation of TgSERCA expression led to a marked defect in replication, with parasites failing to progress beyond one or two rounds of division (**Fig. 3E–F**). All parasitophorous vacuoles (PVs) in ATc-treated cultures contained four or fewer parasites (**Fig. 3F**, *bar graph on the right*). Host cell invasion was also reduced in the *ΔTgSERCA* mutant (**Fig. 3G**) when cultured with ATc.

Parasite egress was significantly affected by TgSERCA depletion. Ionomycin (IO), which has been known to trigger egress by inducing Ca^{2+} release (Borges-Pereira et al., 2015), and natural egress following pre-incubation with 1 μM compound 1 (Donald et al., 2002), previously shown to synchronize parasite exit (Vella et al., 2021), were both markedly reduced in ATc-treated parasites, underscoring the critical role of ER Ca^{2+} stores in supporting both ionophore-induced and spontaneous egress (**Fig. 3H**, *IO and natural egress*). Interestingly, however, egress induced by saponin in the presence of extracellular Ca^{2+} was accelerated in ATc-treated parasites (**Fig. 3H**, *saponin egress*). This enhancement may result from a more rapid rise in cytosolic Ca^{2+} , reaching the egress threshold more quickly due to impaired SERCA activity combined with ongoing Ca^{2+} leak from the ER (Vella et al., 2021). The saponin concentration used selectively permeabilizes the host cell membrane, allowing extracellular Ca^{2+} to enter the parasite cytosol without compromising the integrity of the parasite plasma membrane. This is consistent with previous observations showing that tachyzoites remain motile and exhibit Ca^{2+} oscillations under similar conditions (Borges-Pereira et al., 2015). The resulting rise in cytosolic Ca^{2+} within the parasite stimulates motility and triggers egress. To further examine this phenomenon, we directly compared the timing of egress between untreated and ATc-treated *ΔTgSERCA* parasites under identical saponin exposure conditions (**Fig. 3H**).

In summary, our findings demonstrate that TgSERCA is essential for *T. gondii* replication, invasion, and natural egress. Interestingly, when host cells were selectively permeabilized, parasites with reduced TgSERCA expression displayed accelerated egress, likely due to altered calcium dynamics.

Ca^{2+} uptake by the SERCA-ATPase is essential for filling acidic Ca^{2+} stores

Further characterization of the *ΔTgSERCA* mutant showed a diminished cytosolic Ca^{2+} response to TG (**Fig. 4A**), which was also observed when TG was applied after extracellular Ca^{2+} had been added to fill the stores (**Fig. 4B**). This was most likely due to reduced Ca^{2+} accumulation by the ER in the *ΔTgSERCA* (+ATc) mutant. Note that the change in Ca^{2+} in **Fig. 4A** (*Tati/ku80* cells) is larger than in **Fig. 1B** (RH strain), which we attribute to differences between the two cell lines (RH vs *Tati/ku80*).

Most importantly, MgATP-driven Ca^{2+} uptake by permeabilized cells measured using Mag-Fluo-4, showed no TgSERCA activity after 48 h of culture with ATc (**Fig. 4C**). This experiment validated the Mag-Fluo-4 method for assessing SERCA activity. At 24 hours post-culture with ATc, some residual SERCA activity was still detected (**Fig. 4C**).

Interestingly, Ca^{2+} entry measured in Fura-2 loaded *ΔTgSERCA* parasites ($\pm$ ATc) was not affected by the downregulation of TgSERCA (**Fig. 4D**). This finding argues against the presence of an ER-based mechanism that regulates Ca^{2+} entry. Moreover, the cytosolic resting Ca^{2+} concentration remained unchanged in the *ΔTgSERCA* (+ATc) mutant (**Fig. 4A,B,D–H**) highlighting a critical role of the plasma membrane Ca^{2+} pump in maintaining cytosolic Ca^{2+} homeostasis.

The response to Zaprinast was diminished but was still present (**Fig. 4E**) indicating that Zaprinast induces Ca^{2+} release from the ER and from an additional compartment. When Zaprinast was added after Ca^{2+} replenishment, the response remained reduced in the mutant pre-incubated

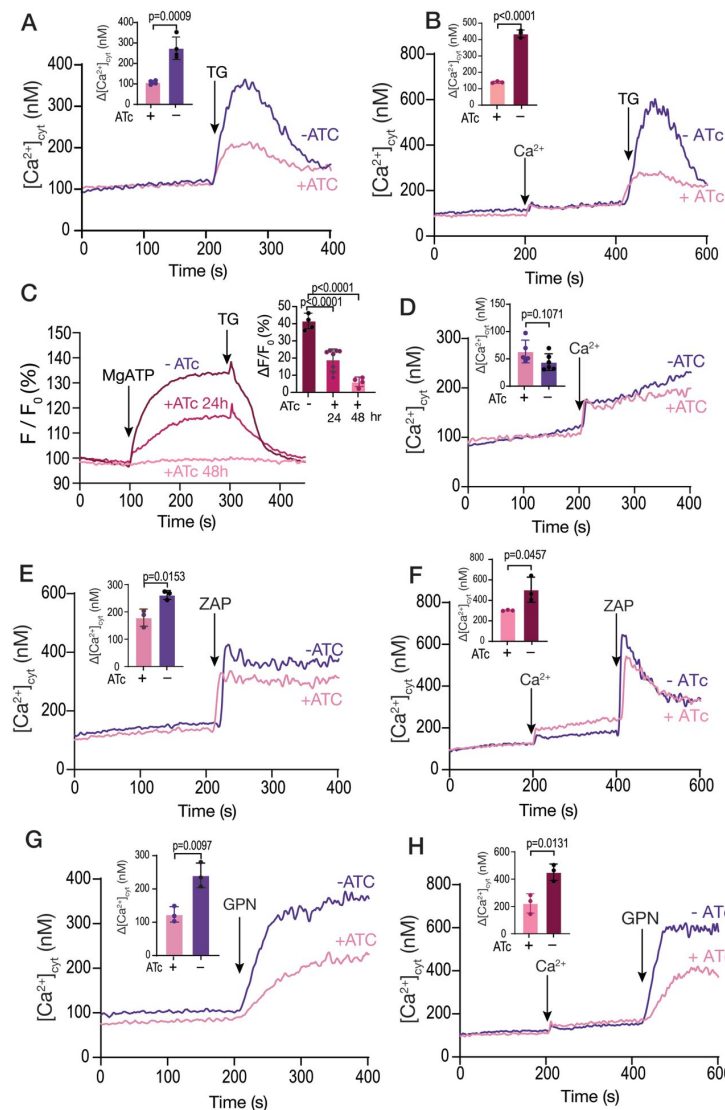


Figure 4

Organellar calcium pools in the $\Delta TgSERCA$ mutant.

A, The $\Delta TgSERCA$ mutant was grown $\pm$ ATc and was loaded with Fura-2 for cytosolic Ca^{2+} measurements. 1 μM TG was added at 200 sec to a suspension of tachyzoites. The purple trace shows the response of the parental cell line grown without ATc and the pink trace shows the response of the same mutant grown with ATc for 24 h. The bar graph shows the analysis of the $\Delta[Ca^{2+}]_{cyt}$ from three biological experiments. B, same experimental set-up as the one in A but adding 1.8 mM extracellular Ca^{2+} at 200 sec. C, SERCA activity measured in Mag-Fluo-4 loaded $\Delta TgSERCA$ tachyzoites grown $\pm$ ATc. Parasites were collected, loaded with Mag-Fluo-4 AM, and permeabilized with digitonin as described in the Methods section. Free Ca^{2+} in the buffer was set at 220 nM, and MgATP (0.125 mM) was added at 100 s. The purple trace represents the control (no ATc), while the other traces correspond to parasites treated with ATc for 24 or 48 h. TG (1 μM) was added as indicated. The bar graph shows the quantification of the initial slope after adding MgATP. D, Ca^{2+} entry measured in Fura-2-loaded $\Delta TgSERCA$ parasites grown $\pm$ ATc. Extracellular Ca^{2+} (1.8 mM) was added at 200 s. The inset shows ΔF values from three independent experiments, indicating no significant differences. E, Similar conditions to the ones used in A but adding 100 μM Zaprinas. The bar graph shows the quantification of the $\Delta[Ca^{2+}]$ from three biological experiments. F, Similar conditions to the ones used in B but adding 1.8 mM extracellular calcium at 200 sec and 100 μM Zaprinas at 400 sec. The bar graph shows the quantification of the $\Delta[Ca^{2+}]$ from 3 biological experiments. G, Same as A but adding 40 μM GPN. The bar graph shows the analysis of the $\Delta[Ca^{2+}]$ from three biological replicates. H, Same set-up as in F but adding 1.8 mM Ca^{2+} at 200 sec followed by 40 μM GPN at 400 sec. The bar graph shows the quantification of the $\Delta[Ca^{2+}]$ from three biological replicates. Data are presented as mean $\pm$ SD. p value: unpaired two tailed t test performed in all comparisons.

with ATc (**Fig. 4F** [↗](#)). We next tested GPN, which primarily targets acidic stores, and observed a decreased response (**Fig. 4G** [↗](#)). Adding, GPN after replenishing the cells with Ca^{2+} resulted in an increased response as we showed in **Fig. 1** [↗](#), but this response was also reduced when the mutant was grown with ATc (**Fig. 4H** [↗](#)).

Given that the Ca^{2+} phenotypes were assessed after 24 hours of ATc treatment, when approximately 50% of TgSERCA activity remains, the response to Zaprinas may still reflect ER involvement, and may not provide definitive evidence for the contribution of an additional Ca^{2+} pool. To further investigate this, we conducted an experiment in which TG was added prior to GPN and Zaprinas. In this setting, GPN significantly reduced the Zaprinas-induced response (**Fig. S2B** [↗](#)). This result suggests that Zaprinas also targets a non-ER Ca^{2+} store, and that this store is likely the same one affected by GPN.

These results support a functional link between the stores targeted by GPN and the ER. Given that SERCA downregulation impaired ER Ca^{2+} storage without affecting cytosolic Ca^{2+} uptake and cytosolic Ca^{2+} levels, the diminished response to GPN suggests that Ca^{2+} released or leaked from the ER is important for refilling the store targeted by GPN.

The mitochondrion takes up Ca^{2+} from the ER and from acidic stores

In mammalian cells, the high concentration of Ca^{2+} in the ER is important for mitochondrial ATP production (Wenzel et al., 2022 [↗](#)). This is because of the close proximity between the ER and mitochondria which allows for the directional flow of Ca^{2+} from the ER to the mitochondria (Gincel et al., 2001 [↗](#); Rapizzi et al., 2002 [↗](#)). With the aim of verifying if the *T. gondii* mitochondria can take up calcium, we introduced a genetic Ca^{2+} indicator in the mitochondrion of *T. gondii* tachyzoites by attaching the *GCaMP6f* gene (Chen et al., 2013 [↗](#)) to the mitochondrial targeting signal of the *T. gondii* superoxide dismutase 2 (SOD2) gene (Pino et al., 2007 [↗](#)) and isolated stable transgenic clones (RH-SOD2-*GCaMP6f*) (Vella et al., 2020 [↗](#)). Fluorescence microscopy of live cells confirmed *GCaMP6f* localization to the mitochondria (**Fig. 5A** [↗](#)). Direct Ca^{2+} uptake was observed in digitonin permeabilized parasites incubated in the presence of increasing concentrations of Ca^{2+} (**Fig. 5B-C** [↗](#)). Although a measurable increase in mitochondrial fluorescence was observed, it required high Ca^{2+} concentrations, indicating that the *T. gondii* mitochondrion can take up Ca^{2+} but do so with very low affinity. These Ca^{2+} levels were significantly higher than the typical cytosolic Ca^{2+} concentrations found in healthy cells (**Fig. 5B-C** [↗](#)).

We hypothesized that the *T. gondii* mitochondrion may take up Ca^{2+} through close membrane contacts with the ER, where localized Ca^{2+} concentrations in microdomains could be significantly higher than in the cytosol, a mechanism previously described in mammalian cells (Rizzuto et al., 1998 [↗](#)). We next loaded the RH-SOD2-*GCaMP6f* mutant with Fura-2 to simultaneously monitor cytosolic and mitochondrial Ca^{2+} in intact parasites. Upon addition of extracellular Ca^{2+} , an increase in cytosolic Ca^{2+} was observed, however mitochondrial *GCaMP6f* fluorescence remained unchanged (**Fig. 5D-E** [↗](#)), suggesting that mitochondria are unable to take up Ca^{2+} at the cytosolic concentrations reached under these conditions. This also validates the proper localization of the indicator, confirming its absence from the cytosol. Addition of TG followed by extracellular Ca^{2+} resulted in a cytosolic Ca^{2+} increase, readily detected in Fura-2-loaded parasites. However, and most importantly, only TG triggered a measurable increase in the mitochondrial *GCaMP6f* signal, whereas a rise in cytosolic Ca^{2+} induced by extracellular Ca^{2+} addition alone did not (**Fig. 5G** [↗](#)). Our interpretation is that TG-induced ER Ca^{2+} leakage led to local accumulation of Ca^{2+} at the cytosolic face of the ER membrane, creating microdomains of high Ca^{2+} concentration sufficient to trigger mitochondrial uptake. Addition of Ca^{2+} after TG resulted in a greater increase in cytosolic Ca^{2+} (**Fig. 5F** [↗](#)) compared to TG alone. However, even under these conditions, no corresponding increase in mitochondrial *GCaMP6f* fluorescence was observed. This further confirms that mitochondria are unable to take up cytosolic Ca^{2+} at these low concentrations.

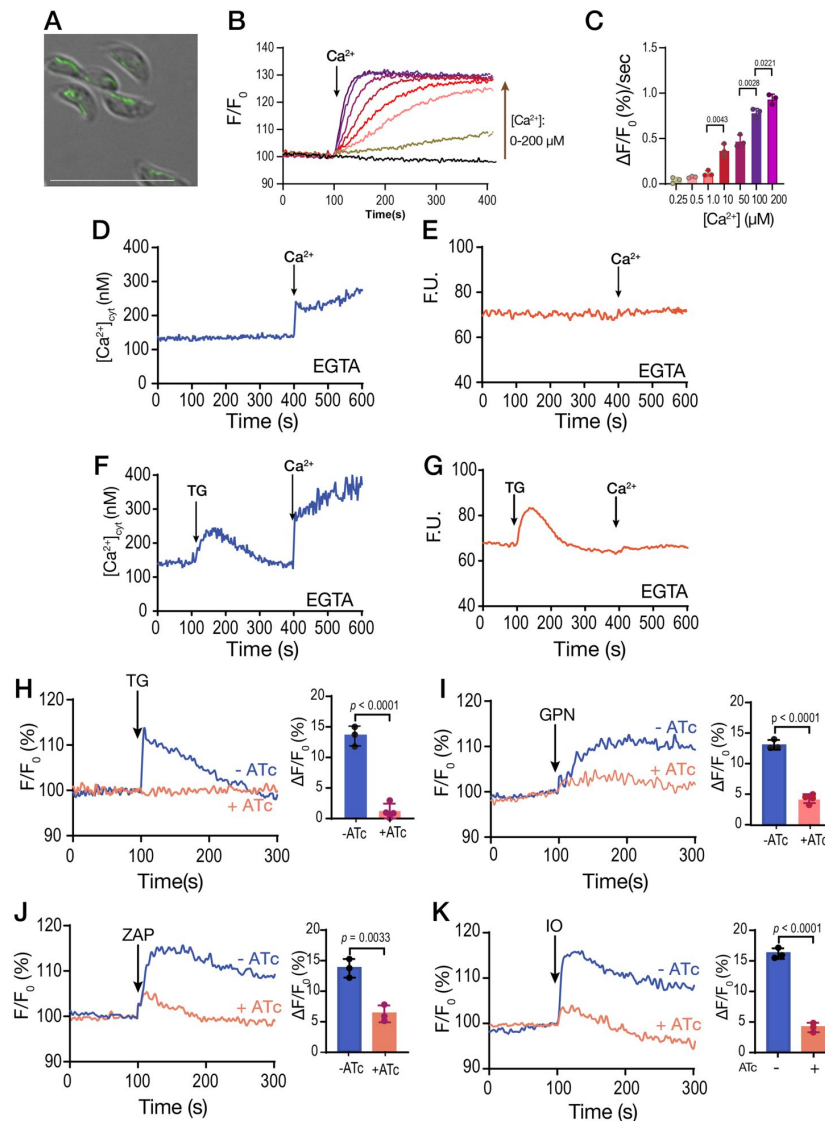


Figure 5

Mitochondrial Calcium uptake.

A, Fluorescence image of *T. gondii* tachyzoites of the RH strain expressing *SOD2-GCaMP6f* (pDT7S4H3-SOD2-GCaMP6f). The generation of this cell line is described in the methods section. **B**, Ca^{2+} uptake in digitonin-permeabilized *T. gondii* tachyzoites expressing *SOD2-GCaMP6f*. Parasites (5×10^7) were permeabilized as described in the Methods section and suspended in buffer containing 100 μM EGTA. Ca^{2+} was added at 100 s to reach final free concentrations of 0.25, 0.5, 1, 10, 50, 100, and 200 μM , calculated using Maxchelator. **C**, ΔF was measured as the change in fluorescence between the baseline and the maximum value obtained 20 s after Ca^{2+} addition. Data represent the average of three independent biological experiments. **D**, Fura-2-loaded *T. gondii* tachyzoites expressing *SOD2-GCaMP6f* in suspension. The experimental setup was identical to that described in **Fig. 1A-B**. CaCl_2 (1.8 mM) was added at 400 s, and fluorescence measurements were performed under Fura-2 conditions. **E**, *GCaMP6f* fluorescence measurements of intact *T. gondii* tachyzoites expressing *SOD2-GCaMP6f* targeted to the mitochondrion. Fluorescence was recorded using optimized settings for *GCaMP6* detection. **F**, 1 μM TG was added at 100 sec followed by 1.8 mM CaCl_2 at 400 sec. Fura-2 loaded parasites and Fura-2 conditions were used. **G**, Same additions as in **F** but measuring fluorescence of *GCaMP6f*. **H**, Response to 1 μM TG of *$\Delta TgSERC$ A-SOD2-GCaMP6f* parasites (transfected with the pCTH3-SOD2-GCaMP6f plasmid), grown with (pink trace) or without (blue trace) ATc. Fluorescence measurements were performed under the same conditions as in panel **G** using intact parasites. The bar graph shows ΔF values from three independent biological replicates. **I**, same as **H** but using 40 μM GPN. **J**, Same as **H** but using 100 μM Zaprinast. **K**, Same as **H** but using 1 μM Ionomycin. Data are presented as mean $\pm$ SD from 3 independent biological experiments. *p* value: unpaired two tailed t test performed in all comparisons.

We next introduced the same *SOD2*-GCaMP6f chimeric gene into the *ΔTgSERCA* mutant background and isolated a clonal line (*ΔTgSERCA-SOD2-GCaMP6f*) (**Fig. S5A-B**). Fluorescence imaging of live cells confirmed proper localization of the indicator (**Fig. S5A**), and fluorescence measurements of intact cells corroborated that adding extracellular Ca^{2+} did not increase GCaMP6f fluorescence, whereas addition of TG to the suspension resulted in a fluorescence increase (**Fig. S5B**). We next monitored changes in GCaMP6f fluorescence in the mutant and compared results between parasites grown with and without ATc. In line with prior observations, parasites cultured without ATc showed a consistent and measurable increase in mitochondrial GCaMP6f signal upon TG treatment (**Fig. 5H**, blue trace). In contrast, this response was abolished in parasites cultured with ATc, consistent with reduced TgSERCA expression leading to ER Ca^{2+} depletion (**Fig. 5H**, pink trace).

We next tested additional stimuli and observed a clear increase in mitochondrial GCaMP6f fluorescence in the *ΔTgSERCA* mutant following the addition of GPN, Zaprinast, or IO (**Fig. 5H-J**, blue traces). In all cases, this fluorescence increase was significantly reduced in the *ΔTgSERCA* (+ATc) mutant (**Fig. 5H-J**, pink traces). These results suggest a potential direct interaction between the mitochondrion and acidic Ca^{2+} stores, such as the PLVAC and/or Golgi apparatus. The reduced Ca^{2+} content of these compartments, resulting from TgSERCA downregulation, appears to impact mitochondrial Ca^{2+} uptake.

In summary, we demonstrated that the *T. gondii* mitochondrion is capable of Ca^{2+} uptake via transfer from the ER, a process that becomes apparent upon inhibition of ER Ca^{2+} uptake with TG. This suggests that the high Ca^{2+} concentrations required for mitochondrial uptake are achieved only at membrane contact sites between the ER and mitochondria. In the *ΔTgSERCA* (+ATc) mutant, impaired ER Ca^{2+} storage due to TgSERCA downregulation compromises mitochondrial Ca^{2+} uptake. Additionally, our data suggest that the *T. gondii* mitochondrion may also take up Ca^{2+} from acidic stores, such as the PLVAC or Golgi, which appear to rely indirectly on ER Ca^{2+} refilling. When TgSERCA is downregulated, depletion of ER Ca^{2+} likely compromises the Ca^{2+} content of these acidic compartments, and impairing mitochondrial Ca^{2+} uptake from these stores.

Proximity between the ER, Mitochondrion and Acidic Compartment

We next investigated if it was possible to see proximity between the ER with other organelles by IFA and/or electron microscopy (EM). We performed IFAs with ER and mitochondria markers and ER and PLVAC markers (**Fig. 6**). In intracellular parasites, the mitochondrion was observed to surround the ER, forming multiple potential sites of interaction (**Fig. 6A** and *supplemental video 1*). As described, the mitochondrion of intracellular *T. gondii* tachyzoites surrounds the periphery of the cell in a lasso-shape morphology (Ovcariikova et al., 2017). In contrast, in extracellular parasites, the mitochondrion displays a marked morphological change, adopting either a sperm-like or collapsed conformation (Ovcariikova et al., 2017) (**Fig. 6B** and *supplemental video 2*). Our hypothesis is that this retraction of the mitochondrion allows the ER membranes to expand in extracellular parasites and extend toward the apical domain of the parasite where Ca^{2+} is needed for secretion of micronemes and conoid extrusion.

The PLVAC also formed points of contact with the ER (**Fig. 6C** and *supplemental video 3*). Multiple points of contact were also observed by EM between the ER and the PLVAC, the ER and the apicoplast and the ER and the mitochondrion (**Fig. 6D**).

Interestingly these contacts were still present in the *ΔTgSERCA* (+ATc) mutant (**Fig. S6, A-D**), as most likely TgSERCA would not be directly involved in the establishment of contacts. We quantified the length of the limiting membrane of the organelle in contact with ER membranes at a distance of less than 30 nm and found that after knockdown of TgSERCA, the contact was not altered. However, Ca^{2+} transported from the ER into the mitochondrion after TG treatment was

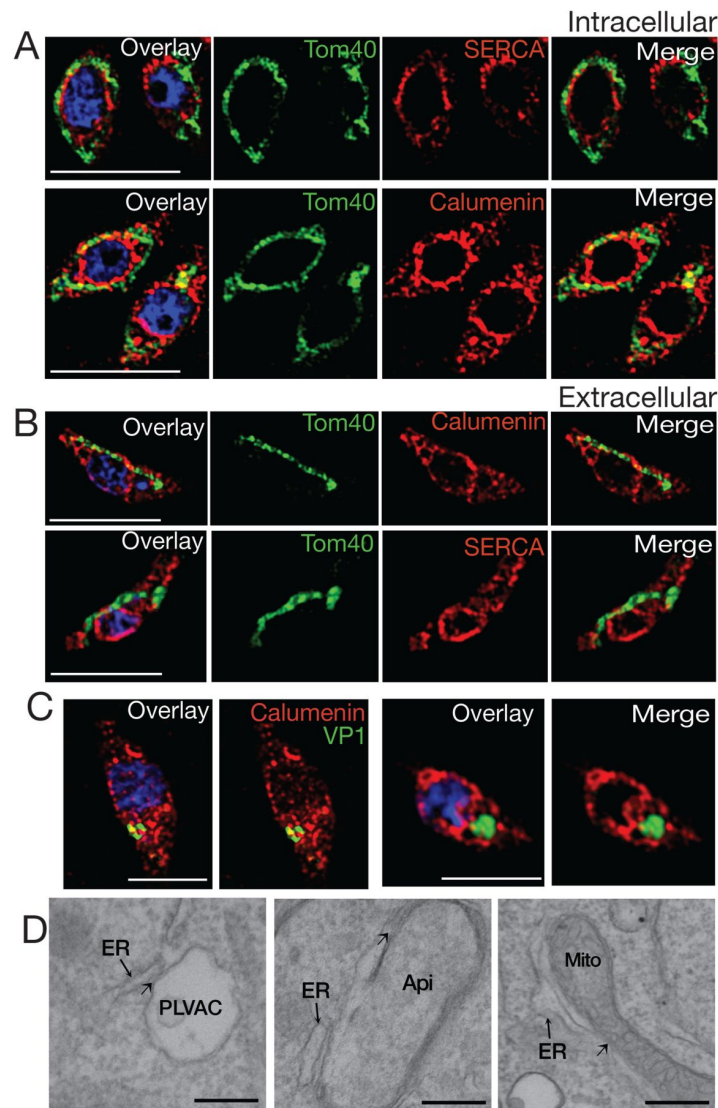


Figure 6

ER membrane contacts with the mitochondrion and the plant like vacuolar compartment (PLVAC).

A, Super-resolution IFAs of intracellular parasites with the mitochondrion labeled with the α Tom40 (green, 1:20,000) antibody and the ER labeled with the α Tgcalumenin antibody (an ER calcium binding protein) (red, 1:1000) or the TgSERCA (red 1:1000). **B**, IFAs of extracellular tachyzoites with the same antibodies used for part A. The mitochondrion and ER membranes interact in several regions. **C**, The PLVAC was labeled with the α TgVP1 antibody (green, 1:200) or the α TgCPL antibody (green, 1:500). The ER was labeled with the α Tgcalumenin antibody (red). The points of contact between the ER and the PLVAC are yellow. Scale bars in A-C is 5 μ m. **D**, Transmission Electron Microscopy imaging of the contact sites formed between ER and PLVAC, ER and Apicoplast, ER and mitochondria. Size bars are 100 nm.

significantly decreased which means that this phenotype is due to reduced ER Ca^{2+} and not to lack of contacts (**Fig. S6E**). A similar result was seen when measuring contacts between the ER and the apicoplast (**Fig. S6F**) and between the ER and the PLVAC (**Fig. S6G**).

In summary, this data supports the presence of points of contact between the ER and other organelles like the mitochondrion, the PLVAC and the apicoplast. These contacts likely facilitate the transfer of Ca^{2+} from the ER, the organelle with the highest Ca^{2+} content, to other compartments.

Discussion

In this work, we demonstrated that the endoplasmic reticulum (ER) of *T. gondii* has a remarkable capacity to sequester Ca^{2+} entering the cytosol from the extracellular milieu, achieving this with only a minimal rise in cytosolic Ca^{2+} levels. This is largely due to the activity of a highly efficient SERCA Ca^{2+} -ATPase (TgSERCA), which has a high affinity for Ca^{2+} . The activity of TgSERCA, most likely together with the plasma membrane Ca^{2+} pump (Luo et al., 2001; Luo et al., 2005), limits large increases in cytosolic Ca^{2+} (Hortua Triana et al., 2024).

We provide evidence that the ER not only sequesters extracellular Ca^{2+} through TgSERCA activity but also shares this pool with other organelles, including mitochondria and acidic stores. This capacity stems from the unique ability of the ER to capture a sizable fraction of extracellular Ca^{2+} entering the tachyzoite cytosol. Such inter-organelle transfer allows localized Ca^{2+} release without globally elevating cytosolic levels, thereby preventing unintended signaling events. Our data support a model in which loss of SERCA activity reduces ER Ca^{2+} as well as Ca^{2+} content in other organelles. Under physiological conditions, ER Ca^{2+} is regularly mobilized for signaling and homeostasis, helping to maintain Ca^{2+} balance across cellular compartments (see our hypothetical model in **Fig. 7**).

SERCA Ca^{2+} -ATPases are P-type pumps located in the ER and secretory pathway membranes (Kuhlbrandt, 2004). Mammals express three isoforms (SERCA1–3), with SERCA2b serving as the housekeeping form (Wuytack et al., 2002). SERCA pumps translocate two Ca^{2+} ions into the ER lumen per ATP hydrolyzed, lowering cytosolic Ca^{2+} to resting levels (<100 nM) and replenishing ER stores (~500 μM). This stored Ca^{2+} supports signaling and the activity of luminal enzymes critical for cell growth, proliferation, and differentiation (Wuytack et al., 2002).

T. gondii appears to express a single SERCA protein (TgSERCA) (Nagamune and Sibley, 2006), likely serving a housekeeping role. The severe defects observed in the *iTgSERCA* (+ATc) mutant like impaired replication and disruption of the lytic cycle, highlight its essential function. TgSERCA activity was dependent on MgATP and exhibited high Ca^{2+} affinity, as evidenced by Mag-Fluo-4-based assays detecting uptake at free Ca^{2+} levels as low as 55 nM. This suggests that TgSERCA functions effectively at physiological cytosolic Ca^{2+} concentrations (60–100 nM), ensuring ER loading even under resting conditions.

In situ characterization of organellar Ca^{2+} uptake has been feasible in trypanosomes (Docampo and Vercesi, 1989; Vercesi et al., 1990) but remains challenging in *T. gondii*. The Mag-Fluo-4 and the mitochondrial GCaMP6f protocols enable reliable measurement of ER and mitochondrial Ca^{2+} uptake, respectively. For ER Ca^{2+} uptake MgATP was essential for the activity of TgSERCA, as other forms of ATP were ineffective. This protocol has been extensively used in mammalian cells, DT40 cells and other cells for measuring intraluminal calcium, activity of SERCA and response to IP3 (Laude et al., 2005; Rossi et al., 2009; Valverde et al., 2010; Sampieri et al., 2018; Rossi and Taylor, 2020). We previously successfully employed it for the characterization of the *Trypanosoma brucei* IP3R (Huang et al., 2013) and the assessment of SERCA activity in *T. gondii* mutants (Li et al., 2021). In this work, we used it to assess TgSERCA activity under defined Ca^{2+} and MgATP conditions.

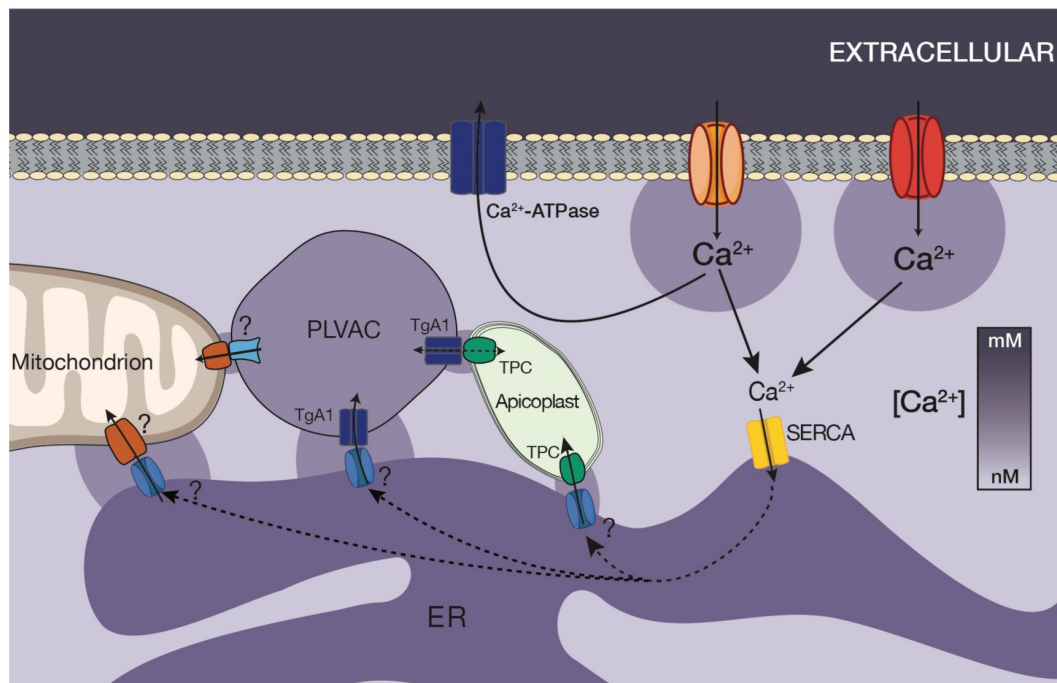


Figure 7

Hypothetical model showing calcium entry through two different types of Ca²⁺ channels, uptake by TgSERCA into the ER and distribution to the other organelles via transfer from the ER to the mitochondria, PLVAC and apicoplast.

The mitochondrion is shown in close contact to the ER which constitutively leaks Ca²⁺ into the cytosol. Ca²⁺ could leak from the ER through the TgTRPPL-2 channel previously described (Marquez-Nogueras et al., 2021 [\[1\]](#)). The mitochondria take up Ca²⁺ from the ER through an unknown mechanism. VDAC could be involved in the transfer through the outer mitochondrial membrane (Mallo et al., 2021 [\[2\]](#)). The PLVAC interacts with the ER and may also interact with the mitochondrion and the apicoplast. TgA1, a calcium ATPase previously characterized may be the pump involved in Ca²⁺ uptake (Luo et al., 2001; Luo et al., 2005). The mechanism of release is unknown. The Two Pore Channel (TgTPC) was shown to be involved in the formation of contacts between the ER and the apicoplast (Li et al., 2021 [\[3\]](#)) and could form part of the mechanism of uptake. Question marks point to molecules or mechanism partially or not yet identified.

Using the Mag-Fluo-4 protocol, we observed that cyclopiazonic acid (CPA), a reversible SERCA inhibitor (Inesi and Sagara, 1994 [\[1\]](#)), induced a Ca^{2+} leak rate comparable to the one after adding TG. This indicates that the leak rate is mainly determined by intrinsic leak mechanisms rather than the type of SERCA inhibition. However, in intact Fura-2-loaded parasites, CPA induced a smaller cytosolic Ca^{2+} increase than TG. This likely reflects CPA's reversible and potentially incomplete inhibition of SERCA under cellular conditions, as was also observed in *Plasmodium falciparum* (Borges-Pereira et al., 2020 [\[2\]](#)).

We observed that the response to acidic calcium triggers like nigericin or GPN were greatly enhanced when added after TG in Fura-2-loaded tachyzoites, likely due to ER Ca^{2+} leak and subsequent transfer to other compartments. Additionally, downregulation of *TgSERCA* expression resulted in reduced responses to these acidic store triggers, supporting the notion that the ER contributes to the filling of these organelles. It is important to note that our analyses of Ca^{2+} storage was done in parasites that retained partial *TgSERCA* activity, as it is not possible to isolate cells entirely lacking *TgSERCA* expression. Under these conditions, Zaprinast still induced a reduced Ca^{2+} mobilization response. This residual response may be due to remaining calcium in the ER or may suggest that Zaprinast targets multiple calcium stores. We recently identified the Golgi apparatus as a calcium store in *T. gondii* (Calixto et al., 2025 [\[3\]](#)) and demonstrated that treatment with GPN in Fura-2-loaded tachyzoites diminished the Zaprinast-induced calcium response, suggesting that Zaprinast and GPN may act on overlapping stores. In the present study, we demonstrated that sequential treatment with TG followed by GPN almost completely abolished the Zaprinast response, further supporting this idea. Although GPN is primarily known to act on acidic organelles, it has also been proposed to affect the ER (Atakpa et al., 2019 [\[4\]](#)) however, we have no evidence that GPN mobilizes calcium from the ER in *T. gondii*. We propose that GPN primarily targets the PLVAC, but further investigation is required to fully characterize its mechanism of action.

It was interesting that Ca^{2+} entry remained unchanged in the *il1TgSERCA* (+ATc) mutant, suggesting that intracellular stores may not be directly involved in the regulation of Ca^{2+} entry (Pace et al., 2014 [\[5\]](#)). Moreover, conventional genomic analysis has not identified homologs of the canonical Store-Operated Calcium Entry (SOCE) components, STIM and Orai (Collins and Meyer, 2011 [\[6\]](#)), raising the possibility that these proteins are either absent or highly divergent in sequence and lack conserved regulatory domains. If communication between intracellular stores and the plasma membrane exists in *T. gondii*, the underlying mechanism remains unclear.

Calcium transport into the *T. gondii* mitochondrion had not been previously demonstrated and our findings provide the first experimental evidence for this process, though the molecular mechanism remains unclear. We found that normal cytosolic Ca^{2+} fluctuations were insufficient to drive mitochondrial uptake, consistent with the low Ca^{2+} affinity of the mitochondrion. Uptake occurred only after SERCA inhibition, which caused local Ca^{2+} accumulation at the cytosolic side of the ER membrane, enabling transfer to the mitochondrion, likely via membrane contact sites (MCSs), since direct uptake from the cytosol would be inefficient at low Ca^{2+} concentrations. MCSs were defined as stable, tightly apposed, but non-fusogenic regions of close proximity between subcellular organelles, and play a key role in inter-organelle communication (Phillips and Voeltz, 2016 [\[7\]](#)). In mammalian cells, the ER forms an extensive network of MCSs with the PM, mitochondria and endocytic vesicles for the exchange of Ca^{2+} (Burgoyne et al., 2015 [\[8\]](#)).

In *T. gondii* the characterization of MCSs is only in its beginnings (Huet and Moreno, 2023 [\[9\]](#)) with only a few evidences for their presence (Tomova et al., 2009 [\[10\]](#); Mallo et al., 2021 [\[11\]](#); Ovciarikova et al., 2022 [\[12\]](#); Ovciarikova et al., 2024 [\[13\]](#)) and function (Li et al., 2021 [\[14\]](#); Oliveira Souza et al., 2022 [\[15\]](#)). Imaging of intracellular *T. gondii* showed that its mitochondrion surrounds the periphery of the cell in a lasso-shape conformation. On the other hand, in extracellular parasites the mitochondrion changes its morphology and adopts a sperm-like or collapsed conformation (Ovciarikova et al., 2017 [\[16\]](#)). Our IFA analysis with ER and mitochondrial markers revealed that the lasso-shaped

mitochondrion surrounds the ER with plenty of opportunities for contact between both organelles. Interestingly, the mitochondrion also appeared capable of importing Ca^{2+} from acidic stores such as the PLVAC, as GPN treatment stimulated mitochondrial Ca^{2+} uptake. This response was reduced in the *iΔTgSERCA* (+ATc) mutant, indicating that TgSERCA activity contributes to this process. Moreover, ER Ca^{2+} depletion impaired transfer from acidic stores to the mitochondrion, suggesting functional interdependence among Ca^{2+} stores and a central role for the ER in coordinating intracellular Ca^{2+} dynamics.

In mammalian cells Ca^{2+} ion is transferred from the ER to the mitochondrion through the outer membrane voltage-dependent anion channel1 (VDAC1) (Gincel et al., 2001 [↗](#); Rapizzi et al., 2002 [↗](#)) and the inner membrane calcium uniporter (MCU1) (De Stefani et al., 2011 [↗](#)). A VDAC homologue is present in *T. gondii*, which was shown to be essential for growth and for mitochondrial and ER morphology (Mallo et al., 2021 [↗](#)). However, molecular evidence for the presence of a Ca^{2+} uniporter in the inner mitochondrial membrane, driven by the electrochemical gradient generated by the electron transport chain, remains to be demonstrated.

Cellular responses triggered by Ca^{2+} signals are shaped by the location, duration, and amplitude of the signals. Movement of Ca^{2+} in the cytosol of cells is severely limited due to the presence of high affinity Ca^{2+} buffers. In mammalian cells it was shown that Ca^{2+} tunnels through the ER as it moves faster because the Ca^{2+} binding capacity of the ER is almost 100 lower than the binding capacity of the cytosol (Mogami et al., 1997 [↗](#)). The ER Ca^{2+} transport through its lumen was shown to provide a mechanism of delivering Ca^{2+} to targeted sites without activating inappropriate processes in the cell cytosol (Petersen et al., 2017 [↗](#)). In *T. gondii*, the relative Ca^{2+} -binding capacity of the cytosol compared to the ER remains poorly understood, as the localization of many predicted Ca^{2+} -binding proteins has not been fully determined. Several calmodulin-like proteins, for example, are localized to the conoid (Long et al., 2017 [↗](#)).

The mechanisms of Ca^{2+} entry at the plasma membrane, release from the ER, and uptake by the mitochondria or acidic stores remain incompletely characterized (Hortua Triana et al., 2018 [↗](#); Pace et al., 2020 [↗](#)) (Garcia et al., 2017 [↗](#)). Consequently, the molecular elements required for classical Ca^{2+} tunneling have not been identified in *T. gondii*. Nevertheless, our results demonstrate that Ca^{2+} can be transferred from the ER to other organelles. This is supported by the increased mitochondrial and acidic calcium pools observed following pharmacological ER depletion, both in the presence and absence of extracellular calcium. Importantly, chronic ER calcium depletion, such as in the *iΔTgSERCA* mutant cultured with ATc, leads to the depletion of all intracellular Ca^{2+} stores. Additionally, we directly demonstrated mitochondrial Ca^{2+} uptake when Ca^{2+} accumulated on the cytosolic side of the ER membrane following SERCA inhibition. The specific roles of Ca^{2+} in the mitochondrion and acidic compartments remain unclear. In mitochondria, Ca^{2+} may support ATP production, although this has yet to be confirmed. Both organelles may also act as auxiliary Ca^{2+} reservoirs during ER Ca^{2+} overload.

In *T. gondii* cytoplasmic Ca^{2+} increases, due to efflux from the ER or entry through the PM, have been reported to initiate key parasite processes such as microneme secretion (Carruthers and Sibley, 1999 [↗](#); Nagamune et al., 2007b [↗](#)), conoid extrusion (Del Carmen et al., 2009 [↗](#)) (Pace et al., 2014 [↗](#)), invasion (Vieira and Moreno, 2000 [↗](#); Lovett and Sibley, 2003 [↗](#)) and egress (Arrizabalaga et al., 2004 [↗](#); Borges-Pereira et al., 2015 [↗](#)). These responses require precise spatiotemporal regulation of Ca^{2+} at specific cellular sites, suggesting the presence of mechanisms that direct Ca^{2+} to discrete locations. We propose that the ER plays a central role in this regulation by acting as a hub that distributes Ca^{2+} to defined sites at defined times to initiate parasite functions. The severe invasion, replication, and egress defects observed in the *iΔTgSERCA* (+ATc) mutant support this hypothesis. Egress was one of the first steps of the *T. gondii* lytic cycle that was shown to be triggered by exposure of intracellular parasites to ionophores (Endo et al., 1982 [↗](#)). Most recent work using GECIs demonstrated the rise in cytosolic calcium preceding egress (Borges-Pereira et al., 2015 [↗](#); Stewart et al., 2017 [↗](#)). In the present study, we demonstrate that

egress is defective and unresponsive to ionophores in parasites lacking sufficient Ca^{2+} in their intracellular stores. This underscores the critical role of TgSERCA in maintaining Ca^{2+} stores filled. Interestingly, this defect could be rescued by exposing the parasites to extracellular calcium through host cell permeabilization, which not only restored egress but also was faster.

In conclusion, this study demonstrates that the ER of *T. gondii* can replenish itself with Ca^{2+} and acts as a source of Ca^{2+} for cytosolic signaling, as well as for loading acidic stores and the mitochondrion. *T. gondii* is a protozoan parasite that causes disease by reiterating a lytic cycle that is driven by Ca^{2+} signaling. Our findings enhance understanding of how extracellular and intracellular Ca^{2+} stores coordinate to sustain the pathologic features of *T. gondii*. Future studies will focus on defining the roles of Ca^{2+} in mitochondrial and acidic stores functions.

Methods

Cell Culture

Toxoplasma gondii tachyzoites (RH and *TatiΔku80* strain) were maintained in human telomerase reverse transcriptase immortalized foreskin fibroblasts (hTERT) (Farwell et al., 2000 [DOI](#)) grown in Dulbecco's modified minimal essential media (DMEM) with 1% FBS. These cells are tested for Mycoplasma contamination regularly and are treated with mycoplasma removal agent.

Generation of SERCA mutants

A promoter insertion plasmid was generated by cloning three PCR fragments into a modified pCR2.1-TOPO vector using Gibson Assembly Cloning Kit (NEB #E5510). One fragment corresponding to the TgSERCA flanking region (predicted promoter/5'UTR) was amplified with primers 1 and 2 (Table S1). The second fragment corresponds to DHFR+T7S4 (Sheiner et al., 2011 [DOI](#)) and was amplified with primers 3 and 4. Another fragment corresponds to the 5' TgSERCA coding sequence beginning with start codon and was amplified with primers 5 and 6. The vector pCR2.1-TOPO was used, which had only one EcoRI site and it was cut with the enzyme NotI to use as a vector backbone. The promoter insertion plasmid was transfected into the *TatiΔku80* cells and selected with 1 μM pyrimethamine using a "ultra aggressive" screening method. Briefly, 200 μl of the suspension of transfected parasites was added to 10 ml of media and then one to three drops (~65 μl per drop) were inoculated into three 24-well plates containing media. The clonal lines created after selection and subcloning were termed *iΔTgSERCA*.

For *in situ* tagging, an approximately 2 kb fragment was amplified from the genomic locus (3' region) of the TgSERCA gene using primers 7 and 8. The fragment was cloned into the pLic-3HA-CAT plasmid (Huynh and Carruthers, 2009 [DOI](#)) and the construct was linearized with the enzyme NheI for transfection into the *iΔTgSERCA* mutant. Clonal cell lines were generated after selection with chloramphenicol and subcloning and termed *iΔTgSERCA-3HA*.

Expression and purification of TgSERCA recombinant protein

The phosphorylation (P) and nucleotide binding(N) domains of TgSERCA (TGGT1_230420) (nucleotides 1123 to 2415, amino acid residues 375 to 805) were cloned into XmaI and HindIII sites of pQE-80L with primers 13 and 14 to create recombinant protein with a N-terminal 6xHis tag. The resulting plasmid was transformed into *Escherichia coli* BL21-CodonPlus competent cells and expression was induced by addition of 0.4 mM isopropyl Δ -D-1-thiogalactopyranoside (IPTG) for 4 hours at 37°C. Cells were pelleted and resuspended in equilibration/binding buffer (50 mM Na_3PO_4 , 300 mM NaCl, 10 mM Imidazole, 8 M Urea, and protease inhibitor cocktail P-8849). The cells were then sonicated for 80 seconds and centrifuged at 12,000 rpm for 20 min at 4°C. The supernatant was filtered through a 0.45 μm membrane and the protein was purified using HisPur Ni-NTA Chromatography Cartridge (Thermo Scientific) following instructions from the

manufacturer. Proteins that were unbound were washed with 12 ml of wash buffer (50 mM Na₃PO₄, 300 mM NaCl, 40 mM imidazole, and 8 M urea), and the recombinant protein was eluted with 5 ml elution buffer (50 mM Na₃PO₄, 300 mM NaCl, 250 mM imidazole, and 8 M urea). Eluted protein fractions were concentrated and desalted with Amicon Ultra-0.5 mL centrifugal filter (Millipore Sigma).

Anti-TgSERCA antibody generation in Guinea pigs

Two Guinea pigs were each immunized with 0.2 mg of purified recombinant protein mixed with equal volume of Freund's Complete Adjuvant (Sigma F5581), followed by two boosts of 0.1 mg antigen mixed with equal volume of Freund's Incomplete Adjuvant (Sigma F5506) for Guinea pig 1 and three boosts for Guinea pig 2. The resulting antibodies were tested at 1:1,000 in western blot against RH lysates and were developed with Alexa Fluor 488 goat anti-Guinea pig (1:1,000). The antibodies were compared with Dr. Sibley's mouse anti-SERCA antibody (Nagamune et al., 2007a [↗](#)) to confirm size and purity. The anti-TgSERCA antibodies were then affinity purified. Guinea pigs were handled according to our approved institutional animal care and use committee (IACUC) protocols (A2021 03-005-A5) of the University of Georgia.

Cytosolic calcium measurements with Fura-2

T. gondii tachyzoites were loaded with Fura-2 AM as previously described (Vella et al., 2020 [↗](#); Stasic et al., 2021 [↗](#)). Freshly released tachyzoites were washed twice with buffer A plus glucose (BAG; 116 mM NaCl, 5.4 mM KCl, 0.8 mM MgSO₄, 50 mM HEPES, pH 7.3, and 5.5 mM glucose), by centrifugation (706 x *g* for 10 min) and re-suspended to a final density of 1 x 10⁹ parasites/ml in loading buffer (BAG plus 1.5% sucrose, and 5 μM Fura-2-AM). The suspension was incubated for 26 min at 26 °C with mild agitation. Subsequently, the parasites were washed twice (2,000 x *g* for 2 min) with BAG to remove extracellular dye, re-suspended to a final density of 1x10⁹ parasites per ml in BAG and kept on ice. This loading protocol is specifically designed to minimize Fura-2 compartmentalization, which is typically indicated by elevated resting Ca²⁺ concentrations. All experiments are conducted within a time frame during which resting Ca²⁺ levels remain stable, typically below or at 100 nM. For fluorescence measurements, 2 x 10⁷ parasites/mL were placed in a cuvette with 2.5 mL of Ringer's buffer without calcium (155 mM NaCl, 3 mM KCl, 1 mM MgCl₂, 3 mM NaH₂PO₄, and 10 mM Hepes, and 10 mM glucose). Fluorescence measurements were done in a Hitachi F-7000 or F-4500 fluorescence spectrophotometer using the Fura-2 conditions for excitation (340 and 380 nm) and emission (510 nm). The Fura-2 fluorescence response to Ca²⁺ was calibrated from the ratio of 340/380 nm fluorescence values after subtraction of the background fluorescence of the cells at 340 and 380 nm as previously described (Gryniewicz et al., 1985 [↗](#)). The Ca²⁺ release rate was defined as the change in Ca²⁺ concentration during the initial 20 s after reagent addition. Delta F was calculated by the difference between the higher calcium peak and basal Ca²⁺, and recovery was defined as the change of Ca²⁺ concentration after the calcium peak was reached for the subsequent indicated times.

Endoplasmic reticulum Ca²⁺ measurements in permeabilized *T. gondii* tachyzoites

Tachyzoites freshly egressed and washed as described above were resuspend to a final density of 1x10⁹ cells/ml in HBS buffer (135 mM NaCl, 5.9 mM KCl, 1.2 mM MgCl₂, 11.6 mM HEPES pH 7.3, 1.5 mM CaCl₂, 11.5 mM glucose) containing 1 mg/ml BSA, 0.2 mg/ml of pluronic F127 and 20 μM Mag-Fluo4-AM. The suspension was incubated at RT with mild shaking for 1 h, in the dark. Subsequently, parasites were washed 2 times at 5,000 rpm for 2 min to remove extracellular dye. The pellet was resuspended in 1.8 ml of CLM buffer (20 mM NaCl, 140 mM KCl, 20 mM PIPES, pH 7.0) containing 1 mM EGTA at 1x10⁹ cells/ml. Cells were permeabilized with 44.4 μg/ml digitonin for 6 min. Cells were washed 2 times with CLM containing 1 mM EGTA at 5,000 rpm for 2 min to remove digitonin and resuspended to a final density of 1x10⁹ tachyzoites/ml and kept in ice. For each test, 50 μl (5x10⁷) of parasite suspension was added to 1.95 ml of CLM containing 1 mM EGTA

and 0.375 mM CaCl₂ which results in 220 nM free Ca²⁺ as calculated with MaxChelator. Fluorescence was measured with a Hitachi F-7000 or F-4500 fluorescence spectrophotometer (Excitation at 485 nm and emission at 520 nm). Ratio ($\Delta F/F_0$ /sec) was evaluated by measuring the rate of change in fluorescence over 20 s after reagent addition.

Strain construction and maintenance

The organelle targeting of GCaMP6f was made by overlapping PCR. The N-terminal mitochondrial targeting sequence of the *T. gondii* SOD2 gene (Pino et al., 2007 [\[1\]](#)) was used to target GCaMP6f to the mitochondrion. The *GCaMP6f* gene for this construct was amplified by primers 9 and 10 (Table S1). After gel purification of the GCaMP6f and SOD2 sequences, the mitochondria targeting construct was built by overlapping PCR with the purified PCR products as template. This construct was then cloned into the Topo-blunt vector. After the sequence was verified by sequencing, the *SOD2-GCaMP6f* fragment was removed by BglII and AvrII digestion and cloned into the same restriction sites of the pDT7S4H3 (Sheiner et al., 2011 [\[2\]](#)) and pCTH3 (Vella et al., 2020 [\[3\]](#)) vectors. The pDT7S4H3-SOD2-GCaMP6f construct was introduced into RH parasites by electroporation. After selection with pyrimethamine, the parasites were sorted by FACS and then subcloned. Clones were selected based on the dynamic range of the response to ionomycin. The pCTH3-SOD2-GCaMP6f was introduced into the *il1TgSERC*A mutant by electroporation. After selection with chloramphenicol, the parasites were sorted by FACS and then subcloned. The expression of GCaMP6f was verified by live cell imaging, IFA, and western blots. The clone with the largest dynamic range, as evaluated with Ionomycin, was selected to further use. Fig. S5 [\[4\]](#) shows IFA and fluorescence controls for mitochondria localization and function of the mitochondrial GCaMP6f.

GCaMP6f fluorescence measurements

Permeabilized tachyzoites: *T. gondii* tachyzoites expressing SOD2-GCaMP6f were collected and washed 2 times at 5,000 rpm for 2 min with BAG. The parasite pellet was resuspended in 1.8 ml of BAG buffer containing 0.1 mM EGTA at 1×10^9 cells/ml. Permeabilization with 44.4 μ g/ml digitonin for 6 min was done by following the fluorescence of GCaMP6f. Cells were washed 2 times with the same buffer at 5,000 rpm for 2 min to remove digitonin, resuspended to a final density of 1×10^9 parasites/ml in intracellular buffer (140 mM Kgluconate, 10 mM NaCl, 2.7 mM MgSO₄, 200 μ M EGTA, 65 μ M CaCl₂, 10 mM HEPES, 10 mM Tris, pH 7.3, 1 mM Glucose) and kept in ice. 50 μ l (5×10^7) of the parasite suspension was mixed with 1.95 ml intracellular buffer for measurement. Measurements were done in a Hitachi 7000 fluorescence spectrophotometer set at 485 nm excitation and 509 nm emission. The uptake rate ($\Delta F/F_0$ /Sec) was evaluated by measuring the % of change in fluorescence per second during the initial 20 s after reagent addition.

For measurements with intact parasites, they were collected, washed, and resuspended in BAG at 1×10^9 cells/ml for testing. Fifty μ l (5×10^7) of the parasite suspension was mixed with 1.95 ml BAG containing 0.1 mM EGTA for measurement. The ratio ($\Delta F/F_0$) was evaluated by measuring the maximum change in fluorescence over 20 s after reagent addition (linear rate).

Growth, invasion and egress assays

Red-green invasion assays were performed as originally described (Kafsack et al., 2004 [\[5\]](#)), modified (Chasen et al., 2017 [\[6\]](#)) and adapted to use td-RFP expressing parasites. The number of tachyzoites used was 2×10^7 , and invasion was for 5 min. Plaque assays were performed as previously described (Roos et al., 1994 [\[7\]](#); Liu et al., 2014 [\[8\]](#)) with modifications (Liu et al., 2014 [\[8\]](#)). 125 tachyzoites were used for infection of confluent six well plates with hTERT fibroblasts followed by an incubation time of 10 days prior to fixing and staining with crystal violet.

For egress assays, the monolayers of hTERT cells grown in 35 mm Mattek dishes were infected with 50,000 tdTomato-expressing parasites for 24 or 48 h. Parasitophorous vacuoles containing 4–8 parasites were observed by microscopy after washing twice with Ringer's buffer without calcium.

The dish was filled with 1 ml of Ringer's buffer supplemented with either 100 μ M EGTA or 1.8 mM CaCl_2 . Images were collected in time-lapse mode with an acquisition rate of 3 seconds for 15 minutes. We observed that most of the *il1TgSERCA* cells +/- ATc were still able to egress when stimulated with 1 or 0.5 μ M ionomycin added 2 min after the start of the recordings with either 100 μ M EGTA or 1.8 mM CaCl_2 . We next used lower concentrations of ionomycin (100 nM and 50 nM) in Ringer's buffer with 1.8 mM CaCl_2 or 0.01% saponin. For egress triggered by ionomycin, the percentage of vacuoles egressed after adding ionomycin during the 15 minutes of the video (2 minutes baseline + 13 minutes after adding ionomycin) was quantified (from 100 vacuoles). For egress triggered by saponin, the time to egress after adding saponin was quantified. For natural egress, the *il1TgSERCA* mutant expressing td-tomato RFP was used to infect confluent hTERT cell monolayers 36 hours before adding ATc and 1 μ M compound 1 (pyrrole 4- [2-(4-fluorophenyl)-5-(1-methylpiperidine-4-yl)-1H-pyrrol-3-yl]pyridine) (Cpd1) (Donald et al., 2002 [DOI](#)) dissolved in ethanol and the culture continued for 24 hours. After treating with Cpd1 for 24 hours, cultures showed full vacuoles which differed from the vehicle treated plates (36 hours cultured plus ATc treatment for 24 hours or 48 hours without ATc), which were fully lysed. Following synchronization, the Cpd1-containing media was removed, and the vacuoles were washed twice with warm media lacking Cpd1. Fresh media without Cpd1 was added, and the plates were transferred to a prewarmed DeltaVision microscope stage set to 37 °C. After 10 min at 37°C, egress of the full vacuoles was enumerated. We counted each plate for 1 min and we counted at least 100 vacuoles per experiment. Three independent biological experiments were conducted and summarized.

For replication assays, hTERT cells were grown on 35 mm Mattek dishes. Each dish was infected with 50,000 tdTomato-expressing parasites. 24 h after the infection counting of parasites per PV was done in a fluorescence microscope. For each experiment, at least 100 PVs were counted. Results were the average of 3 independent experiments (Li et al., 2021 [DOI](#)).

Microscopy and western blot analyses

Tachyzoites were grown on hTERT cells on cover slips for ~24 hr, washed twice with BAG and fixed with 4% formaldehyde for 1 h, followed by permeabilization with 0.3% Triton X-100 for 20 min, and blocking with 3% bovine serum albumin. IFAs were performed as previously described (Miranda et al., 2010 [DOI](#)). Fluorescence images were collected with an Olympus IX-71 inverted fluorescence microscope with a Photometrix CoolSnapHQ CCD camera driven by DeltaVision software (Applied Precision, Seattle, WA). Super-resolution microscopy was performed using a Zeiss ELYRA S1 (SR-SIM) system mounted on a high-resolution Axio Observer Z1 inverted microscope. The setup included transmitted light (HAL), UV (HBO), and high-power solid-state laser illumination sources (405/488/561 nm), a 100 $\times$ oil immersion objective, and an Andor iXon EM-CCD camera. Image acquisition and structured illumination analysis were conducted using ZEN software (Zeiss) with the SIM analysis module. Rat anti-HA antibody (Roche) was used at a 1:25 dilution, and mouse anti-HA antibody (Covance) was used at a 1:200 dilution. Affinity-purified guinea pig anti-TgSERCA antibody was used at a 1:500 dilution.

Western blot analysis was performed as previously described (Liu et al., 2014 [DOI](#)). Rat anti-HA antibody from Roche was used at a dilution of 1:200. Mouse anti-HA antibody from Covance was used at a dilution of 1:1,000. The guinea pig anti-TgSERCA antibody was used at a dilution of 1:2,000. Secondary goat anti-rat or mouse antibody conjugated with HRP was used at 1:5,000. Mouse anti- α -tubulin at a dilution of 1:5,000 was used for loading control.

Transmission Electron Microscopy

For ultrastructural observations of intracellular *T. gondii* by thin-section transmission electron microscopy (EM), infected human foreskin fibroblast cells were fixed in 2.5% glutaraldehyde in 0.1 mM sodium cacodylate (EMS) and processed as described (Coppens and Joiner, 2003 [DOI](#)). Ultrathin sections of infected host cells were stained before examination with a Hitachi 7600 EM under 80 kV. For quantitative measurement of distance between organelles, the closest point between *T.*

gondii's organelles and ER membrane was measured using ImageJ and was performed on 47 representative electron micrographs at the same magnification for accurate comparison between organelles

Statistical Analysis

Statistical analyses were performed by Student's t-test using GraphPad PRISM version 9. Error bars shown represent mean $\pm$ SD (standard deviation) of at least three independent biological replicates. Unpaired two tailed t test performed in all comparisons.

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Additional files

Supplemental tables and Figures [🔗](#)

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Author information

Zhu-Hong Li

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States

Beejan Asady

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States, Department of Molecular Microbiology and Immunology, Johns Hopkins Bloomberg School of Public Health, Baltimore, United States

Le Chang

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States

Miryam Andrea Hortua Triana

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States

Catherine Li

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States

Isabelle Coppens

Department of Molecular Microbiology and Immunology, Johns Hopkins Bloomberg School of Public Health, Baltimore, United States

Silvia NJ Moreno

Center for Tropical and Emerging Global Diseases, Department of Computer Science, University of Georgia, Athens, United States, Department of Cellular Biology, University of Georgia, Athens, United States
ORCID iD: [0000-0002-2041-6295](https://orcid.org/0000-0002-2041-6295)

For correspondence: smoreno@uga.edu

Editors

Reviewing Editor

Malcolm McConville

The University of Melbourne, Melbourne, Australia

Senior Editor

Kenton Swartz

National Institute of Neurological Disorders and Stroke, Bethesda, United States of America

Reviewer #1 (Public review):

Li et al. investigate Ca^{2+} signaling in *T. gondii* and argue that Ca^{2+} tunnels through the ER to other organelles to fuel multiple aspects of *T. gondii* biology. They focus in particular on TgSERCA as the presumed primary mechanism for ER Ca^{2+} filling. Although, when TgSERCA was knocked out there was still a Ca^{2+} release in response to TG present. Overall the data supports a model where the Ca^{2+} filling state of the ER modulates Ca^{2+} dynamics in other organelles.

Comments on revisions:

I thank the authors for their careful revisions and response to my comments, which have been addressed.

Regarding the most critical point of the paper that is Ca^{2+} transfer from the ER to other organelles, the authors in their rebuttal and in the revised manuscript argue that ER Ca^{2+} is critical to redistribute and replenish Ca^{2+} in other organelles in the cell. I agree this conclusion and think it is best stated in the authors' response to point #7: "We propose that this leaked calcium is subsequently taken up by other intracellular compartments. This effect is observed immediately upon TG addition. However, pre-incubation with TG or knockdown of SERCA reduces calcium storage in the ER, thereby diminishing the transfer of calcium to other stores."

In their rebuttal the authors particularly highlight experiments in Figures 1H-K, 4G-H, and 5H-K in support of this conclusion. The data in Fig 1H-K show that with TG there is increased Ca^{2+} release from acidic stores. In all cases TG results in a rise in cytoplasmic Ca^{2+} that could load the acidic stores. So under those conditions the increased acidic organelle Ca^{2+} is likely due to a preceding high cytosolic Ca^{2+} transient due to TG. The experiments in 4G-H and 5H-K are more convincing and supportive of an important role of ER Ca^{2+} to maintain Ca^{2+} levels in other organelles. Overall, and to avoid a detailed, lengthy discussion of every point, the data support a model where in the absence of SERCA activity ER Ca^{2+} is reduced as well as Ca^{2+} in other organelles. I think it would be helpful to present and discuss this finding throughout the manuscript as under physiological conditions ER Ca^{2+} is regularly mobilized for signaling and homeostasis and this maintains Ca^{2+} levels in other organelles. This is supported by the new experiment in Supp Fig. 2A.

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Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Li et al. investigate Ca²⁺ signaling in T. gondii and argue that Ca²⁺ tunnels through the ER to other organelles to fuel multiple aspects of T. gondii biology. They focus in particular on TgSERCA as the presumed primary mechanism for ER Ca²⁺ filling. Although, when TgSERCA was knocked out there was still a Ca²⁺ release in response to TG present.

Note that we did not generate a complete SERCA knockout, as this gene is essential, and its complete loss would not permit the isolation of viable parasites. Instead, we created conditional mutants that downregulate the expression of SERCA. Importantly, some residual activity is present in the mutant after 24 h of ATc treatment as shown in Fig 4C. This is consistent with our Western blots, which demonstrate the presence of residual SERCA protein at 1, 1.5 and 2 days post ATc treatment (Fig. 3B). We have clarified this point in the revised manuscript (lines 232233). See also lines 97-102.

Overall the Ca²⁺ signaling data do not support the conclusion of Ca²⁺ tunneling through the ER to other organelles in fact they argue for direct Ca²⁺ uptake from the cytosol. The authors show EM membrane contact sites between the ER and other organelles, so Ca²⁺ released by the ER could presumably be taken up by other organelles but that is not ER Ca²⁺ tunneling. They clearly show that SERCA is required for T. gondii function.

Overall, the data presented to not fully support the conclusions reached

We agree that the data does not support Ca²⁺ tunneling as defined and characterized in mammalian cells. In response to this comment, we have modified the title and the text accordingly.

However, we respectfully would like to emphasize that the study demonstrates more than just the role of SERCA in T. gondii “function”. Our findings reveal that the ER, through SERCA activity, sequesters calcium following influx through the PM (see reviewer 2 comment). The ER calcium pool is important for replenishing other intracellular compartments.

The experiments support a model in which the ER actively takes up cytosolic Ca²⁺ as it enters the parasite and contributes to intracellular Ca²⁺ redistribution during transitions between distinct extracellular calcium environments. We believe that the role of the ER in modulating intracellular calcium dynamics is demonstrated in Figures 1H–K, 4G–H, and 5H–K. To highlight the relevance of these findings, we have included an expanded discussion in the revised manuscript. See lines 443-449 and 510-522.

Data argue for direct Ca²⁺ uptake from the cytosol

The ER most likely takes up calcium from the cytosol following its entry through the PM and redistributes it to the other organelles. We deleted any mention of the word “tunneling” and replaced it with transfer and re-distribution as they reflect our experimental findings more accurately.

We interpret the experiments shown in Figure 1 H and I as re-distribution because the amount of calcium released after nigericin or GPN are greatly enhanced after TG addition. We first add calcium to allow intracellular stores to become filled, followed by the addition of TG, which allows calcium leakage from the ER. This leaked calcium can either enter the cytosol and be pumped out or be taken up by other organelles. Our interpretation is that this process leads to an increased calcium content in acidic compartments.

We conducted an additional experiment in which SERCA was inhibited prior to calcium addition, allowing cytosolic calcium to be exported or taken up by acidic stores. We observed a change in the GPN response (Fig. S2A), possibly indicating that the PLVAC can sequester calcium when SERCA is inactive. While this may support the reviewer's view, TG treatment does not reflect physiological conditions and may enhance calcium transfer to other compartments. Although the result is interesting, interpretation is complicated by the use of parasites in suspension and drug exposure in solution. Single-parasite measurements are not feasible due to weak signals, and adhered parasites are even less physiological than those in suspension.

In support of our view, the experiments shown in Figs 4G and H show that down regulating SERCA reduces significantly the response to GPN indicating diminished acidic store loading. In Fig 5I we observe that mitochondrial calcium uptake is reduced in the iDSERCA (+ATC) mutant in response to GPN. Fig 2B demonstrates that TgSERCA can take up calcium at 55 nM, close to resting cytosolic calcium while in Figures 5E and S5B we show that the mitochondrion is not responsive to an increase of cytosolic calcium. Uptake by the mitochondria requires much higher concentrations (Fig 5B-C), which may be achieved within microdomains at MCS between the ER and mitochondrion. This is also consistent with findings reported by Li et al (Nat Commun. 2021) where similar microdomains mediated transfer of calcium to the apicoplast (Fig. 7 E and F of the mentioned reference) was observed.

Reviewer 2 (Public review):

The role of the endoplasmic reticulum (ER) calcium pump TgSERCA in sequestering and redistributing calcium to other intracellular organelles following influx at the plasma membrane.

T. gondii transitions through life cycle stages within and exterior to the host cells, with very different exposures to calcium, adds significance to the current investigation of the role of the ER in redistributing calcium following exposure to physiological levels of extracellular calcium

They also use a conditional knockout of TgSERCA to investigate its role in ER calcium store-filling and the ability of other subcellular organelles to sequester and release calcium. These knockout experiments provide important evidence that ER calcium uptake plays a significant role in maintaining the filling state of other intracellular compartments.

We thank the reviewer.

While it is clearly demonstrated, and not surprising, that the addition of 1.8 mM extracellular CaCl₂ to intact T. gondii parasites preincubated with EGTA leads to an increase in cytosolic calcium and subsequent enhanced loading of the ER and other intracellular compartments, there is a caveat to the quantitation of these increases in calcium loading. The authors rely on the amplitude of cytosolic free calcium increases in response to thapsigargin, GPN, nigericin, and CCCP, all measured with fura2. This likely overestimates the changes in calcium pool sizes because the buffering of free calcium in the cytosol is nonlinear, and fura2 (with a K_d of 100-200 nM) is a substantial, if not predominant, cytosolic calcium buffer. Indeed, the increases in signal noise at higher cytosolic calcium levels (e.g. peak calcium in Figure 1C) are indicative of fura2 ratio calculations approaching saturation of the indicator dye.

We acknowledge the limitations associated with using Fura-2 for cytosolic calcium measurements. However, according to the literature (Grynkiewicz, Get al. (1985). J. Biol.

Chem. 260 (6): 3440–3450. PMID 3838314) Fura-2 is suited for measurements between 100 nM and 1 μ M calcium. The responses in our experiments were within that range and the experiments with the SERCA mutant and mitochondrial GCaMPs supports the conclusions of our work.

However, we agree with the reviewer that the experiment shown in Fig 1C (now Fig 1D) presents a response that approaches the limit of the linear range of Fura-2. In response to this, we have replaced this panel with a more representative experiment that remains within the linear range of the indicator (revised Fig 1D). Additionally, we have included new experiments adding GPN along with corresponding quantifications, which further support our conclusions regarding calcium dynamics in the parasite.

Another caveat, not addressed, is that loading of fura2/AM can result in compartmentalized fura2, which might modify free calcium levels and calcium storage capacity in intracellular organelles.

We are aware of the potential issue of Fura-2 compartmentalization, and our protocol was designed to minimize this effect. We load cells with Fura-2 for 26 min at room temperature, then maintain them on ice, and restrict the use of loaded parasites to 2-3 hours. We have observed evidence of compartmentalization as this is reflected in increasing concentrations of resting calcium with time. We carry out experiments within a time frame in which the resting calcium stays within the 100 nM range. We have included a sentence in the Materials and Methods section. Lines 604-606.

Additionally, following this reviewer's suggestion, we performed further experiments to directly assess compartmentalization. See below the full response to reviewer 2.

The finding that the SERCA inhibitor cyclopiazonic acid (CPA) only mobilizes a fraction of the thapsigargin-sensitive calcium stores in T. gondii coincides with previously published work in another apicomplexan parasite, P. falciparum, showing that thapsigargin mobilizes calcium from both CPA-sensitive and CPA-insensitive calcium pools (Borges-Pereira et al., 2020, DOI: 10.1074/jbc.RA120.014906). It would be valuable to determine whether this reflects the off-target effects of thapsigargin or the differential sensitivity of TgSERCA to the two inhibitors.

This is an interesting observation, and we now include a discussion of this result considering the Plasmodium study and include the citation. Lines 436-442.

Figure S1 suggests differential sensitivity, and it shows that thapsigargin mobilizes calcium from both CPA-sensitive and CPA-insensitive calcium pools in T. gondii. Also important is that we used 1 μ M TG as we are aware that TG has shown off-target effects at higher concentrations. TG is a well-characterized, irreversible SERCA inhibitor that ensures complete and sustained inhibition of SERCA activity. In contrast, CPA is a reversible inhibitor whose effectiveness is influenced by ATP levels, and it may only partially inhibit SERCA or dissociate over time, allowing residual Ca^{2+} reuptake into the ER.

Additionally, as suggested by the reviewer we performed experiments using the Mag-Fluo-4 protocol to compare the inhibitory effects of CPA and TG. These results are presented in Fig. S3 (Lines 217-223). Under the conditions of the Mag-Fluo-4 assay with digitonin-permeabilized cells, both TG and CPA showed similar rates of Ca^{2+} leakage following the addition of the inhibitor. This may indicate that under the conditions of the Mag-Fluo-4 experiments the rate of Ca^{2+} leak is mostly determined by the intrinsic leak mechanism and not by the nature of the inhibitor. By contrast, in intact Fura-2-loaded cells, CPA induces a smaller cytosolic Ca^{2+} increase than TG, consistent with less efficient SERCA inhibition likely due to its reversibility and possibly incomplete inhibition under cellular conditions.

The authors interpret the residual calcium mobilization response to Zaprinas observed after ATc knockdown of TgSERCA (Figures 4E, 4F) as indicative of a target calcium pool in addition to the ER. While this may well be correct, it appears from the description of this experiment that it was carried out using the same conditions as Figure 4A where TgSERCA activity was only reduced by about 50%.

We partially agree with the reviewer that 50% knockdown of TgSERCA means that the ER may still be targeted by zaprinast, and that there is no definitive evidence of the involvement of another calcium pool. The Mag-Fluo-4 experiment, while we acknowledge that the fluorescence of MagFluo-4 is not linear to calcium, indicates that SERCA activity is present even after 24 hr of ATc treatment. However, when Zaprinas is added after TG, we observed a significant calcium release in wild type cells. This result suggests the presence of another large calcium pool than the one mobilized by TG (PMID: 2693306).

We recently published work describing the Golgi as a calcium store in Toxoplasma (PMID: 40043955) and we showed in Fig. S4 D-G of that work, that GPN treatment of tachyzoites loaded with Fura-2 diminished the Zaprinas response indicating that they could be impacting a similar store. In the present study we performed additional experiments in which TG was followed by GPN and Zaprinas showing a similar pattern. GPN significantly diminished the Zaprinas response. These results are shown now in Figure S2B. We address these possibilities in the discussion and interpretation of the result. Lines 451-460.

The data in Figures 4A vs 4G and Figures 4B vs 4H indicate that the size of the response to GPN is similar to that with thapsigargin in both the presence and absence of extracellular calcium. This raises the question of whether GPN is only releasing calcium from acidic compartments or whether it acts on the ER calcium stores, as previously suggested by Atakpa et al. 2019 DOI: 10.1242/jcs.223883. Nonetheless, Figure 1H shows that there is a robust calcium response to GPN after the addition of thapsigargin.

The results of the indicated experiments did not exclude the possibility that GPN can also mobilize some calcium from the ER besides acidic organelles. We don't have any evidence to support that GPN can mobilize calcium from the ER either. Based on our unpublished work, we think GPN mainly release calcium from the PLVAC. We included the mentioned citation and discuss the result considering the possibility that GPN may be acting on more than one store. Lines 451-460.

An important advance in the current work is the use of state-of-the-art approaches with targeted genetically encoded calcium indicators (GECIs) to monitor calcium in important subcellular compartments. The authors have previously done this with the apicoplast, but now add the mitochondria to their repertoire. Despite the absence of a canonical mitochondrial calcium uniporter (MCU) in the Toxoplasma genome, the authors demonstrate the ability of T. gondii mitochondrial to accumulate calcium, albeit at high calcium concentrations. Although the calcium concentrations here are higher than needed for mammalian mitochondrial calcium uptake, there too calcium uptake requires calcium levels higher than those typically attained in the bulk cytosolic compartment. And just like in mammalian mitochondria, the current work shows that ER calcium release can elicit mitochondrial calcium loading even when other sources of elevated cytosolic calcium are ineffective, suggesting a role for ER-mitochondrial membrane contact sites. With these new tools in hand, it will be of great value to elucidate the bioenergetics and transport pathways associated with mitochondrial calcium accumulation in T. gondii.

We thank this reviewer praising our work. Studies of bioenergetics and transport pathways associated with mitochondrial calcium accumulation is part of our future plans mentioned in lines 520-522 and 545.

The current studies of calcium pools and their interactions with the ER and dependence on SERCA activity in T. gondii are complemented by super-resolution microscopy and electron microscopy that do indeed demonstrate the presence of close appositions between the ER and other organelles (see also videos). Thus, the work presented provides good evidence for the ER acting as the orchestrating organelle delivering calcium to other subcellular compartments through contact sites in T. gondii, as has become increasingly clear from work in other organisms.

Thank you

Reviewer #3 (Public review):

This manuscript describes an investigation of how intracellular calcium stores are regulated and provides evidence that is in line with the role of the SERCA-Ca²⁺-ATPase in this important homeostasis pathway. Calcium uptake by mitochondria is further investigated and the authors suggest that ER-mitochondria membrane contact sites may be involved in mediating this, as demonstrated in other organisms.

The significance of the findings is in shedding light on key elements within the mechanism of calcium storage and regulation/homeostasis in the medically important parasite Toxoplasma gondii whose ability to infect and cause disease critically relies on calcium signalling. An important strength is that despite its importance, calcium homeostasis in Toxoplasma is understudied and not well understood.

We agree with the reviewer. Thank you

A difficulty in the field, and a weakness of the work, is that following calcium in the cell is technically challenging and thus requires reliance on artificial conditions. In this context, the main weakness of the manuscript is the extrapolation of data. The language used could be more careful, especially considering that the way to measure the ER calcium is highly artificial - for example utilising permeabilization and over-loading the experiment with calcium. Measures are also indirect - for example, when the response to ionomycin treatment was not fully in line with the suggested model the authors hypothesise that the result is likely affected by other storage, but there is no direct support for that.

The Mag-Fluo-4-based protocol for measuring intraluminal calcium is well established and has been extensively used in mammalian cells, DT40 cells and other cells for measuring intraluminal calcium, activity of SERCA and response to IP₃ (Some examples: PMID: 32179239, PMID: 15963563, PMID: 19668195, PMID: 30185837, PMID: 19920131).

Furthermore, we have successfully employed this protocol in previous work, including the characterization of the Trypanosoma brucei IP₃R (PMID: 23319604) and the assessment of SERCA activity in Toxoplasma (PMID: 40043955 and 34608145). The citation PMID: 32179239 provides a detailed description of the protocol, including references to its prior use. In addition, the schematic at the top of Figure 2 summarizes the experimental workflow, reinforcing that the protocol follows established methodologies. We included more references and an expanded discussion, lines 425-435.

We respectfully disagree with the concern regarding potential calcium overloading. The cells used in our assays were permeabilized, which is a critical step that allows to precisely control calcium concentrations. All experiments were conducted at 220 nM free calcium, a

concentration within the physiological range of cytosolic calcium fluctuations. This concentration was consistently used across all studies described above. Importantly, permeabilization ensures that the dye present in the cytosol becomes diluted, and allows MgATP (which cannot cross intact membranes) to access the ER membrane, in addition to be able to expose the ER to precise calcium concentrations.

The Mag-Fluo-4 loading conditions are designed to allow compartmentalization of the indicator to all intracellular compartments and the calcium uptake stimulated by MgATP exclusively occurs in the compartment occupied by SERCA as only SERCA is responsive to MgATP-dependent transport in this experimental setup

Regarding the use of IO, we would like to clarify that its broad-spectrum activity is well documented. As a calcium ionophore, IO facilitates calcium release across multiple membranes, and not just the ER leading to a more substantial calcium release compared to the more selective effect of TG. The results observed with IO were consistent with this expected broader activity and support our interpretation.

Lastly, we emphasize that the experiment in Figure 2 was designed specifically to assess SERCA activity in situ under defined conditions. It was not intended to provide a comprehensive characterization of the role of TgSERCA in the parasite. We now clarify this distinction in the revised Discussion lines 425-435.

Below we provide some suggestions to improve controls, however, even with those included, we would still be in favour of revising the language and trying to avoid making strong and definitive conclusions. For example, in the discussion perhaps replace "showed" with "provide evidence that are consistent with..."; replace or remove words like "efficiently" and "impressive"; revise the definitive language used in the last few lines of the abstract (lines 13-17); etc. Importantly we recommend reconsidering whether the data is sufficiently direct and unambiguous to justify the model proposed in Figure 7 (we are in favour of removing this figure at this early point of our understanding of the calcium dynamic between organelles in Toxoplasma).

We thank the reviewer for the suggestions and we modified the language as suggested. We limited the use of the word "showed" to references to previously published work. We deleted the other words

Figure 7 is intended as a conceptual model to summarize our proposed pathways, and, like all models, it represents a working hypothesis that may not fully capture the complexity of calcium dynamics in the parasite. In light of the reviewer's comments, we revised the figure and legend to clearly distinguish between pathways for which there is experimental evidence from those that are hypothetical.

Another important weakness is poor referencing of previous work in the field. Lines 248-250 read almost as if the authors originally hypothesised the idea that calcium is shuttled between ER and mitochondria via membrane contact sites (MCS) - but there is extensive literature on other eukaryotes which should be first cited and discussed in this context. Likewise, the discussion of MCS in Toxoplasma does not include the body of work already published on this parasite by several groups. It is informative to discuss observations in light of what is already known.

The sentence in which we state the hypothesis about the calcium transfer refers specifically to Toxoplasma. To clarify this, we have now added the phrase "In mammalian cells" (Line 311) and included additional citations, as suggested by the reviewer. While only a few studies have described membrane contact sites (MCSs) in Toxoplasma, we do cite several pertinent articles (e.g., lines 479-486). We believe that we cited all articles mentioning MCS in T. gondii

However, we must clarify to the reviewer that the primary focus of our study is not to characterize or confirm the presence of MCSs in *T. gondii*, but rather to demonstrate functional calcium transfer between the ER and mitochondria. Our data support the conclusion that this transfer requires close apposition of these organelles, consistent with the presence of MCSs.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Line 45: change influx to release as Ca^{2+} influx usually referred to Ca^{2+} entry from the extracellular space. Same for line 71.

Corrected, line 47 and 73

(2) Line 54: consider toning down the strong statement of 'widely' accepted as ER Ca^{2+} subdomain heterogeneity remains somewhat debated.

Changed the sentence to "it has been proposed", Line 56

(3) Line 119-21: A lower release in response to TG is typical and does not reflect TG specific for SERCA. It is due to the slow kinetics of Ca^{2+} leak out of the ER allowing other buffering and transport mechanisms to act. Also, could be a reflection of the duration after TG treatment to allow complete store depletion. Figure S1A-B shows that there is still Ca^{2+} in the stores following TG but the TG signal does not go back to baseline arguing that the leak is still active. Hence the current data does not address the specificity of TG for TgSERCA. Please revise the statement accordingly.

Thank for the suggestion, we changed the sentence to this: "This result could reflect the slow kinetics of Ca^{2+} leak from the ER, allowing other buffering and transport mechanisms to mitigate the phenomenon. Alternatively, it may indicate the duration after TG treatment allowing time to complete store depletion. As shown in Figure S1A-B, residual Ca^{2+} remains in the stores after TG treatment, and the TG-induced phenomenon does not return to baseline, suggesting that the leak remains active". Lines 124-128

(4) Figure 1C: the authors interpret the data 'This Ca^{2+} influx appeared to be immediately taken up by the ER as the response to TG was much greater in parasites previously exposed to extracellular Ca^{2+} '. I don't understand this interpretation, in Ca^{2+} -containing solution it would expected to have a larger signal as TG is likely to activate store-operated Ca^{2+} entry which would contribute to a larger cytosolic Ca^{2+} transient. Does *T. gondii* have SOCE? It cannot be uptake into the ER as SERCA is blocked. Unless the authors are arguing for another ER Ca^{2+} uptake pathway? But why are Ca^{2+} uptake in the ER would lower the signal whereas the data show an increased signal?

We pre-incubated the suspension with calcium to allow filling of the stores, while SERCA is still active, and added thapsigargin (TG) at 400 seconds to measure calcium release. The experiment was designed to introduce the concept that the ER may have access to extracellular calcium, a phenomenon not yet clearly demonstrated in *Toxoplasma*. We did not expect to have less release by TG but if the ER is not efficient in filling after extracellular calcium entry it would be expected to have a similar response to TG. Yes, it is very possible that when we add TG we are also seeing more calcium entry through the PM as we previously proposed that the increased cytosolic Ca^{2+} may regulate Ca^{2+} entry. However, the evidence does not support that this increased entry would be triggered by store depletion. The experiments with the SERCA mutant (Fig. 4D) shows that in the conditional knockout mutant,

the ER is partially depleted, yet this does not lead to enhanced calcium entry, suggesting that the depletion alone is not sufficient to trigger increased influx.

There is no experimental evidence supporting the regulation of calcium entry by store depletion in *Toxoplasma* (PMID: 24867952). We revised the text to clarify this point and expanded the discussion on store-operated calcium entry (SOCE). While it is possible that a channel similar to Orai exists in *Toxoplasma*, it is highly unlikely to be regulated by store depletion, as there is no gene homologous to STIM. If store-regulated calcium entry does occur in *Toxoplasma*, it is likely mediated through a different, still unidentified, mechanism. Lines 461-467.

(5) The choice of adding Ca²⁺ first followed by TG is curious as it is more difficult to interpret. Would be more informative to add TG, allow the leak to complete, and then add Ca²⁺ which would allow temporal separation between Ca²⁺ release from stores and Ca²⁺ influx from the extracellular space. Was this experiment done? If not would be useful to have the data.

Yes, this experiment was already published: PMID: 24867952 and PMID: 38382669.

It mainly highlighted that increased cytosolic calcium may regulate calcium entry most likely through a TRP channel. See our response to point 4 and the description of the new Fig. S2 in the response to point 7.

(6) Line 136-39: these experiments as designed - partly because of the issues discussed above - do not address the ability of organelles to access extracellular Ca²⁺ or the state of refilling of intracellular Ca²⁺ stores. They can simply be interpreted as the different agents (TG, Nig, GPN, CCCP) inducing various levels of Ca²⁺ influx.

Concerning TG, the experiment shown in Fig. 4D shows that depletion of the ER calcium does not result in stimulation of calcium entry, indicating the absence of classical SOCE activation in *Toxoplasma*.

To our knowledge, neither mitochondria nor lysosomes (or other acidic compartments) are capable of triggering classical SOCE in mammalian cells.

Given that the ER in *Toxoplasma* lacks the canonical components required to initiate SOCE, it is unclear why the mitochondria or acidic compartments would be able to do so. While it is possible that *T. gondii* utilizes an alternative mechanism for store-operated calcium entry, investigating such a pathway would require a comprehensive study. In mammalian systems, it took almost 15 years and the efforts of multiple research groups to identify the molecular components of SOCE. Expecting this complex question to be resolved within the scope of a single study is unrealistic.

Our current data show that the mitochondrion is unable to access calcium from the cytosol, as shown in Figure 5E. Performing a similar experiment for the PLVAC would be ideal; however, expression of fluorescent calcium indicators in this organelle has not been successful. This is likely due to the presence of several proteases that degrade expressed proteins, as well as the acidic environment, which quenches fluorescence. These challenges have made studying calcium dynamics in the PLVAC particularly difficult.

To address the reviewer's comment, we performed an additional experiment presented in Fig. S2A. In this experiment, we first inhibited SERCA with thapsigargin (TG), preventing calcium uptake into the ER, and subsequently added calcium to the suspension. Under these conditions, calcium cannot be sequestered by the ER. We then applied GPN and quantified the response, comparing it to a similar experimental condition without TG. Indeed, under these conditions, we observed a significant but modest increase in the GPN-induced response,

suggesting that the PLVAC may be capable of directly taking up calcium from the cytosol. However, this occurs under conditions of SERCA inhibition which creates nonphysiological conditions with elevated cytosolic calcium levels and the presence of TG may promote additional ER leakage, both of which could artificially enhance PLVAC uptake. Under physiological conditions, with functional SERCA activity, the ER would likely sequester cytosolic calcium more efficiently, thereby limiting calcium availability for PLVAC direct uptake. Thus, while the result is intriguing, it may not reflect calcium handling under normal cellular conditions. See lines 172-178.

(7) Figure 1H-I: I disagree with the authors' interpretation of the results (lines 144-153). The data argue that by blocking ER Ca²⁺ uptake by TG, other organelles take up Ca²⁺ from the cytosol where it accumulates due to the leak and Ca²⁺ influx as is evident from the data allowing more release. The data does not argue for ER Ca²⁺ tunneling to other organelles. Tunneling would be reduced in the presence of TG (see PMID: 30046136, 24867608).

We partially agree with this concern. In our experiments, TG was used to inhibit SERCA and block calcium uptake into the ER, allowing calcium to leak into the cytosol. We propose that this leaked calcium is subsequently taken up by other intracellular compartments. This effect is observed immediately upon TG addition. However, pre-incubation with TG or knockdown of SERCA reduces calcium storage in the ER, thereby diminishing the transfer of calcium to other stores.

To further support our claim, we performed additional experiments in the absence of extracellular calcium, now presented in Figure 1J-K. We observed that calcium release triggered by GPN or nigericin was significantly enhanced when both agents were added after TG. These results suggest that calcium initially released from the ER can be sequestered by other compartments. As mentioned, we deleted any mention of “tunneling,” but we believe the data support the occurrence of calcium transfer. New results described in lines 166-171.

The experiment in Fig S2A described in the response to (6) also addresses this concern. Under physiological conditions with functional SERCA, cytosolic calcium would likely be rapidly sequestered by the ER, limiting its availability to other compartments. See lines 172-178.

(8) Line 175: SERCA-dependent Ca²⁺ uptake is higher at 880 nM as would be expected yet the authors state that it's optimal at 220 nM Ca²⁺ ?

Yes, it is true that the SERCA-dependent Ca²⁺ uptake rate is higher at elevated Ca²⁺ concentrations. We chose to use 220 nM free calcium because of several reasons: 1) this concentration is close to physiological cytosolic levels fluctuations; 2) it is commonly used in studies of mammalian SERCA; and 3) calcium uptake is readily detectable at this level. While this may not represent the maximal activity conditions for SERCA, we believe it is a reasonable and physiologically relevant choice for assessing calcium transport activity SERCA-dependent. We added one sentence to the results explaining this reasoning (lines 204-207) and we deleted the word optimal.

(9) Figure 3H: the saponin egress data support the conclusion that organelles Ca²⁺ take up cytosolic Ca²⁺ directly without the need for ER tunneling.

The saponin concentration used permeabilizes the host cell membrane, allowing the intracellular tachyzoite to be surrounded with the added higher extracellular calcium concentration. The saponin concentration used does not affect the tachyzoite membrane as the parasite is still moving and calcium oscillations were clearly seen under similar conditions (PMID: 26374900). The resulting calcium increase in the tachyzoite cytosol is what stimulates parasite motility and egress. Since SERCA activity is reduced in the mutant,

cytosolic calcium accumulates more rapidly, reaching the threshold for egress sooner and thereby accelerating parasite exit. The result does not support that the other stores contribute to this because of the Ionomycin response, which shows that egress is diminished in the mutant, likely because the calcium stores are depleted. We added an explanation in the results, lines 262-269 and the discussion, lines 532-539.

(10) Figure S2: the HA and SERCA signals do not match perfectly? Could this reflect issues with HA tagging, potentially off-target effects? Was this tested?

These are not off-target effects, as we did not observe them in the control cells lacking HA tagging. The HA signal also disappeared after treatment with ATc, further confirming that the IFA signal is specific. We agree with the reviewer that the signals do not align perfectly. This discrepancy could be due to differences in antibody accessibility or the fact that the two antibodies recognize different regions of the protein. We added a sentence about this in the result; lines 240-243.

Reviewer #2 (Recommendations for the authors):

The description of the data of Figures 1B and S1A starting on line 108 would be easier to follow if Figure S1A was actually incorporated into Figure 1. It is not clear why these two complementary experiments were separated since they are both equally important in understanding and interpreting the data.

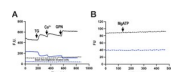
We re-arranged figure 1 and incorporated S1A now as Fig 1C.

As noted in the public comments, loading of fura2/AM can result in compartmentalized fura2, which can contaminate the cytosolic calcium measurements and might modify free calcium levels and calcium storage capacity in intracellular organelles. This can be assessed using the digitonin permeabilization method used in the MagFluo4 measurements, but in this case, detecting the fura2 signal remaining after cell permeabilization.

As suggested by the reviewer, we measured Fura-2 compartmentalization by permeabilizing cells with digitonin as we do for the Mag-Fluo-4 and the fluorescence was reduced almost completely and was unresponsive to any additions (see Author response image 1).

Author response image 1.

T. gondii tachyzoites in suspension exposed to Thapsigargin Calcium and GPN. The dashed lines shows and experiments using the same conditions but parasites were permeabilized with digitonin shows a similar experiment with parasites exposed to MgATP.to release the cytosolic Fura. Part B



Following the public comment regarding the residual calcium mobilization response to Zaprinas observed after 24 h ATc knockdown of SERCA (Figures 4E, 4F, as explained in the legend to Figure 4), was there still a response to Zaprinas after 48 h knockdown, where the thapsigargin response was apparently fully ablated?

Unfortunately, we were unable to perform this experiment as it is not possible to obtain sufficient cells at 48 h with ATc. Due to the essential role of TgSERCA, parasites are unable to replicate after 24 h.

As noted in the public comments, the data in Figure 4A vs 4G and Figure 4B vs 4H appear to show that the calcium responses to GPN are similar to that with thapsigargin, which seems unexpected if the acidic compartment is loaded from the ER. The results with GPN addition after thapsigargin (Figure 1H) argue against this, but the authors should still cite the work of Atakpa et al.

We think that the reviewer is concerned that GPN may also be acting on the ER. This is a possibility that we considered, and we now included the suggested citation (line 457). However, we believe that it is difficult to directly compare the responses, as the kinetics of calcium release from the ER may differ from those of release from the PLVAC. This could be due to differences in the calcium buffering capacity between the two compartments. Additionally, it is possible that calcium leaked from the ER is more efficiently sequestered by other stores or extruded through the plasma membrane than calcium released from the PLVAC. Besides, GPN is known to have a more disruptive effect on membranes compared to TG, which may also influence their responses. As noted by the reviewer, Figure 1H also supports the idea that the acidic compartment is loaded from the ER.

The abbreviation for the plant-like vacuolar compartment (PLVAC) only appears in a figure legend but should be defined in the main text on first use.

Corrected, lanes 140-143

The authors should cite the previous study of Borges-Pereira et al., 2020 (PMID: 32848018) that also demonstrates the incomplete overlap of the calcium pools mobilized by thapsigargin and CPA in *P. falciparum*. The ability to measure calcium in intracellular stores using MagFluo4 opens the possibility to further investigate this discrepancy between CPA and thapsigargin, but CPA does not appear to have been used in the permeabilized cell experiments with MagFluo4. I would suggest that this could be added to Figure 2 and/or Figure 4, or at least as a supplementary figure.

In response to this reviewer's critique we performed additional experiments with Mag-Fluo4 loaded parasites. These are presented in the new Figure S3. We added CPA and TG and combined them to inhibit SERCA and to allow calcium leak from the loaded organelle. Under these conditions, we observed a very similar leak rate after the addition of the inhibitors as measured by the slope of Ca^{2+} leak. We believe that the leak rate is most likely determined by the intrinsic ER mechanism. See the discussion of this result in lines 436442 and the previous response to the same reviewer comment.

Reviewer #3 (Recommendations for the authors):

Suggestions for improved or additional experiments, data, or analyses

(1) Figure 1A is not mentioned in the main text even though it is discussed.

Corrected

(2) Figure 1G: Values do not match, how can GPN be so high?

These figures were replaced by new traces and individual quantification analyses for each experiment.

(3) Figure 1H and I: Is this type of data/results also available for the mitochondrion?

Unfortunately, we were not able to include this experiment because we were unable to accurately quantify the mitochondrial calcium release. Instead, we used mitochondrial GECIs and the results are shown in Figure 5 to study mitochondrial calcium uptake.

(4) Figure 1H: where does the calcium go after GPN addition? Taken up by another calcium store?

Most likely calcium is extruded through the plasma membrane by the activity of the Calcium ATPase TgA1.

However, the reviewer's suggestion is also possible, and calcium could be taken by another store like the mitochondrion. In this regard, we did observe a large mitochondrial calcium increase (parasites expressing SOD2-GCaMp6) after adding GPN (Fig 5I) suggesting that the mitochondrion may take calcium from the organelle targeted by GPN. However, the calcium affinity of the mitochondrion is very low, so the concentration of calcium needs to be very high to activate it and these concentrations are most likely achieved at the microdomains formed between the mitochondrion and other organelles.

(5) Figure 2B-C: Further explanation of why these particular values were chosen for the follow-up experiments would be helpful for the reader.

We tested a wide range of MgATP and free calcium concentrations to measure ER Ca^{2+} uptake catalyzed by TgSERCA. The concentrations shown fall within the linear range.

We followed the free calcium concentrations used by studies of mammalian SERCA (<https://doi.org/10.1016/j.ceca.2020.102188>). In this protocol they used 220 nM free calcium, which was close to cytosolic Ca^{2+} levels. TgSERCA can take up calcium efficiently at this concentration, as shown in Fig 2. We used less MgATP than the mammalian cell protocols, since we did not observe a significant increase in SERCA activity beyond 0.5 mM MgATP. We added one more sentence explaining in the results, lines 204-207.

(6) Figure 3E: Revise the error bar? (and note that colours do not match the graph legend).

The colors do match; the problem visualizing it is because vacuoles containing a single parasite are virtually absent in the control group without ATc treatment.

(7) Figure 3H: 'Interestingly, when testing egress after the addition of saponin in the presence of extracellular Ca^{2+} , we observed that the tachyzoites egressed sooner (Figure 3H, saponin egress).' This is the only graph showing egress timing, and thus it is not clear what is the comparison. The egressed here is sooner compared to what condition? Egress in the absence of Ca^{2+} ? This requires clarification and might require the control data to be added.

In the saponin experiment we compare time to egress of the mutant grown with or without ATc. The measurement is for time to egress after adding saponin. This experiment is in the presence of extracellular calcium. The protocol was previously used to measure time to egress: PMID: 40043955, PMID: 38382669, PMID: 26374900. See also response to question 9 of reviewer 1.

(8) Figure 4C: There is a small peak appearing right after TG addition this should be discussed and explained.

This trace was generated in a different fluorometer, F-4000. This was an artifact due to jumping of the signal when adding TG. Multiple repeats of the same experiment in the newer F7000 did not show the peak. We included in the MM the use of the F-4000 fluorometer for some experiments. We apologize for the omission. Lines 609-610

(9) Figure 5A: An important control that is missing is co-localisation with a mitochondrial marker.

The expression of the SOD2-GCaMP6 has been characterized: PMID: 31758454

(10) Figure 5H: This line was made for this study however the line genetic verification is missing.

In response to this concern we now include a new Figure S5 showing the fluorescence of GCaMP6 in the mitochondrion of the iDTgSERCA mutant (Fig. S5A). We include several parasites. In addition, we show fluorescence measurements after addition of Calcium showing that the cells are unresponsive indicating that the indicator is not in the cytosol. Lines 650-651 and 344-348.

(11) Figure 6D: since the membranes are hard to see, it is not clear whether the arrows show structures that are in line with the definition of membrane contact sites. The authors should provide an in-depth analysis of the length of the interaction between the membranes where the distance is less than 30 nM, and discuss how many structures corresponding to the definition were analysed.

All the requested details are now included in the legend to Figure S3.

Minor corrections to the text and figures

*(1) Unify statistical labelling throughout the paper replacing *** with p values.*

Corrected. We changed the *** with the actual p value in some figures. For figure 2 and Fig S1, we still use the *** due to the space limitation.

(2) Unify ATC vs ATc throughout the paper.

Corrected

(3) Unify capitalization of line name (iΔTgserca/i ΔTgSERCA) throughout the paper.

Corrected

(4) Unify capitalization of p value (p/P) throughout the paper.

Corrected in figures

(5) Unify Fig X vs Fig. X throughout the text.

Corrected

(6) Add values of scale bars to legends (eg Figure S2).

Corrected

| (7) What is the time point for the data in Figures 4E-H, 5H, and S3? 24hrs? include in the legend.

Added 24 h to the legends. Fig S3 is now S4.

| (8) Figure 3F: The second graph is NS thus perhaps no need for the p-value?

Corrected

| (8) Figure 3G: Worth considering swapping the two around: first attachment and then invasion?

Corrected. Invasion and attachment bars were swapped.

| (10) Figure 4A/B: Wrong colour match for Figure 4B.

Corrected

| (11) Figure 4F: In the main text, the authors reference to Figure 1F, correct to 4F.

Corrected

| (12) Figure 4H: In the main text, authors reference to Figure 1H, correct to 4H.

Corrected

<https://doi.org/10.7554/eLife.101894.2.sa0>